



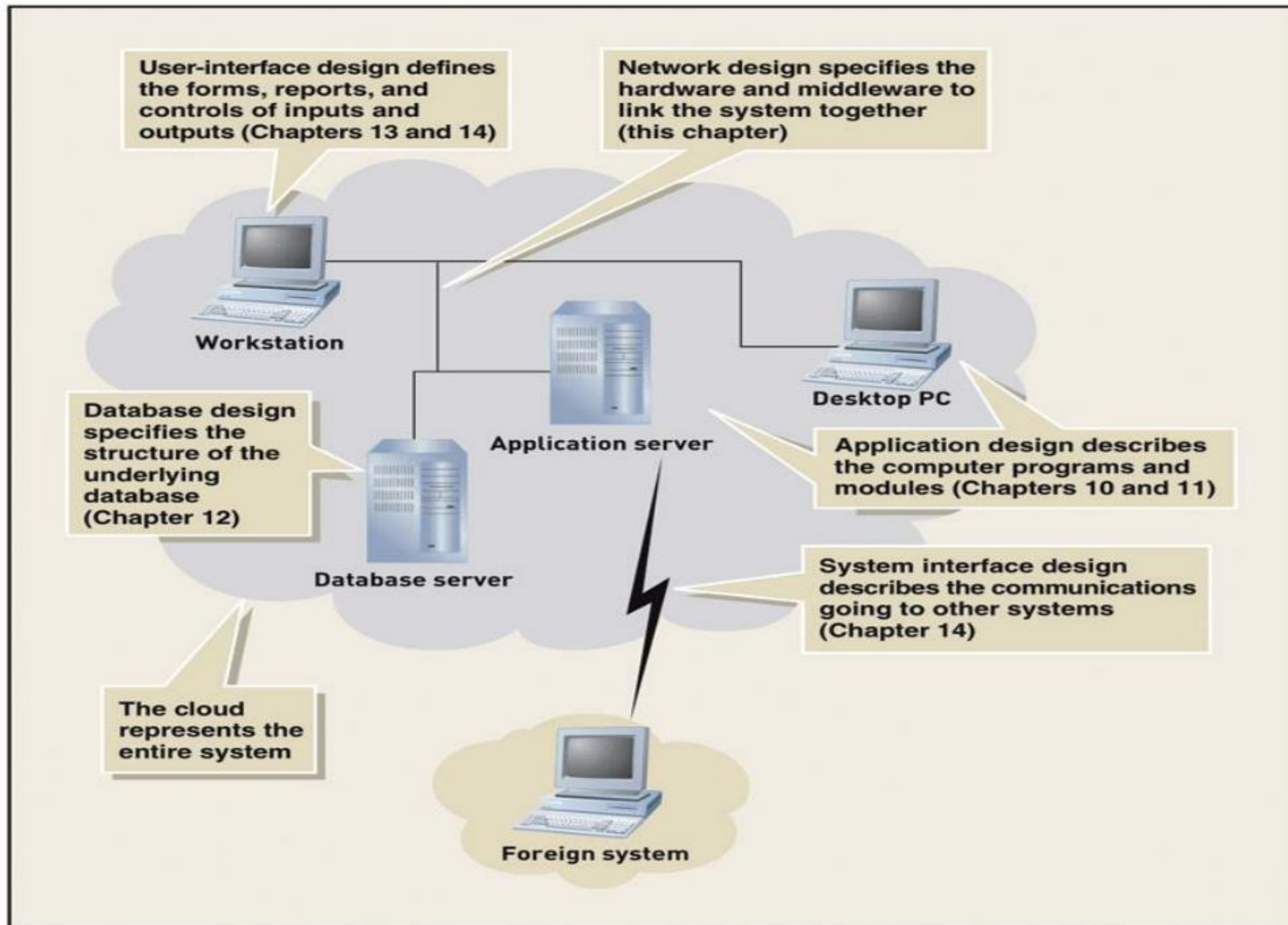
Overview of system design



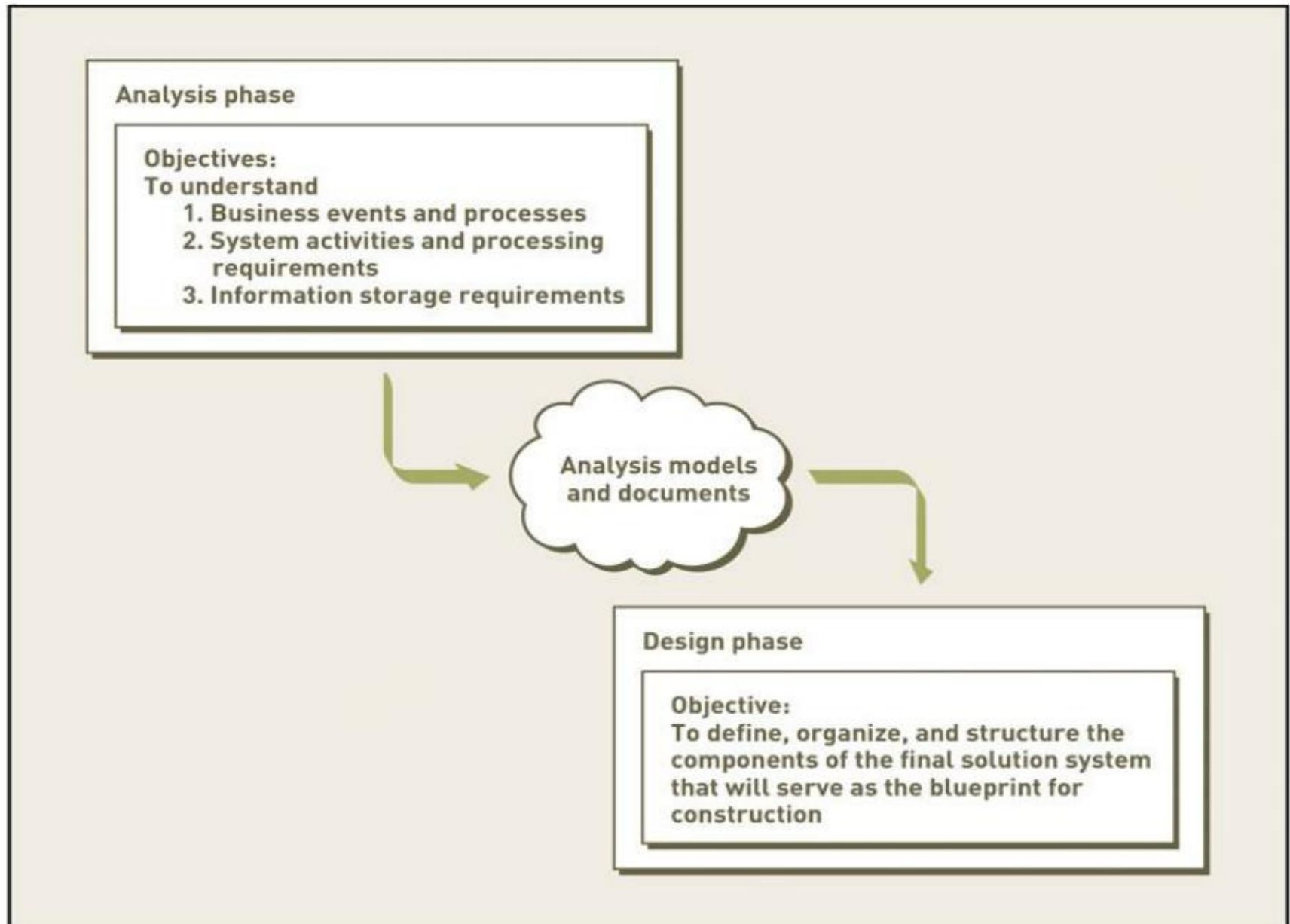
Understanding the Elements of Design

- Design is process of describing, organizing, and structuring system components at **architectural design** level and **detailed design** level
 - Focused on preparing for construction
 - Like developing blueprints
- Three questions
 - What components require systems design?
 - What are inputs to and outputs of design process?
 - How is systems design done?

Components Requiring Systems Design



Analysis Objectives to Design Objectives

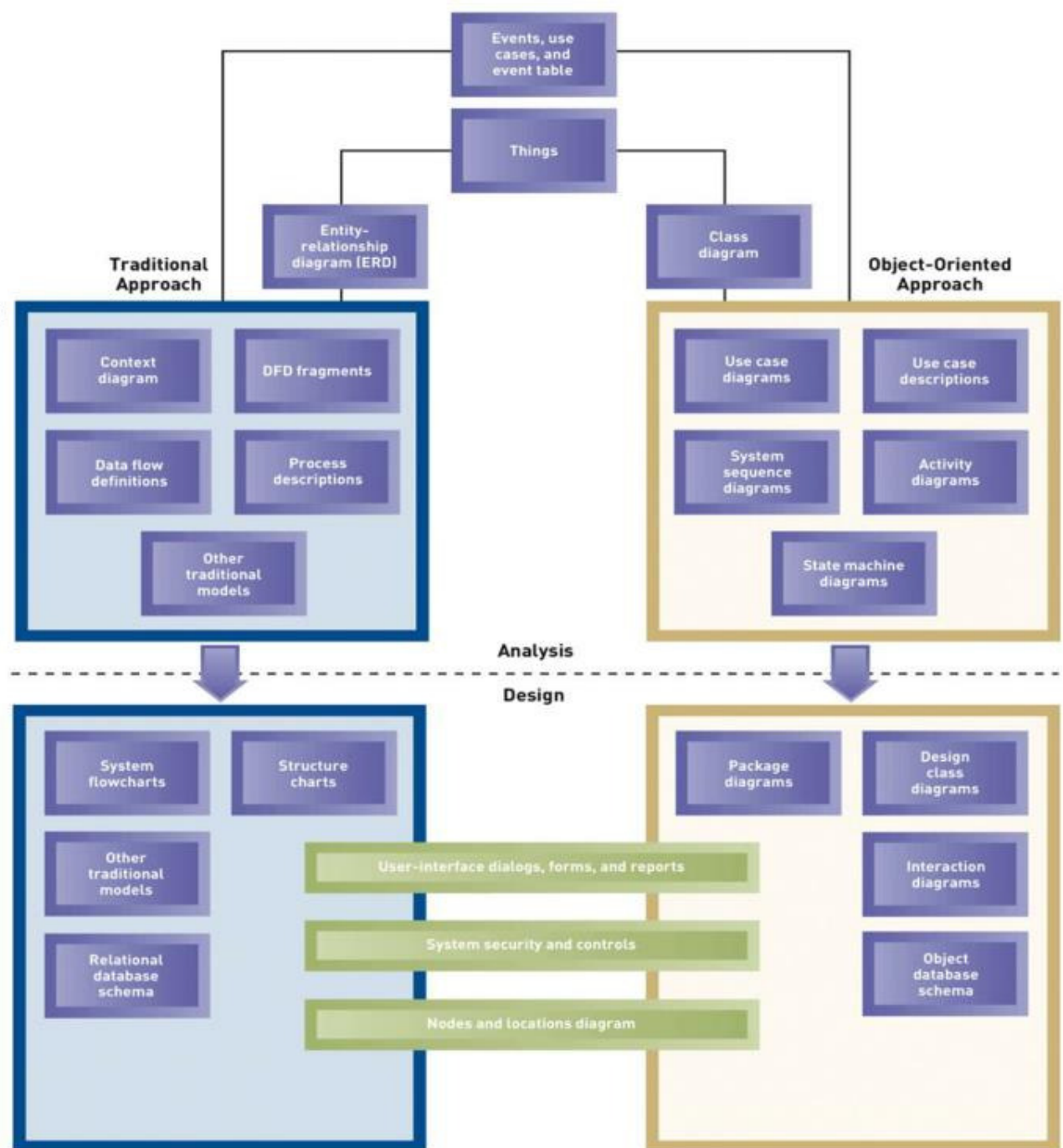




Moving from Analysis to Design

- Design
 - Converts functional models from analysis into models that represent the solution
 - Focused on technical issues
 - Requires less user involvement than analysis
- Design may use structured or OO approaches
 - Database can be relational, OO, or hybrid
 - User interface issues

Traditional Structured and Object-Oriented Models



SDLC Phases with Design Phase Activities

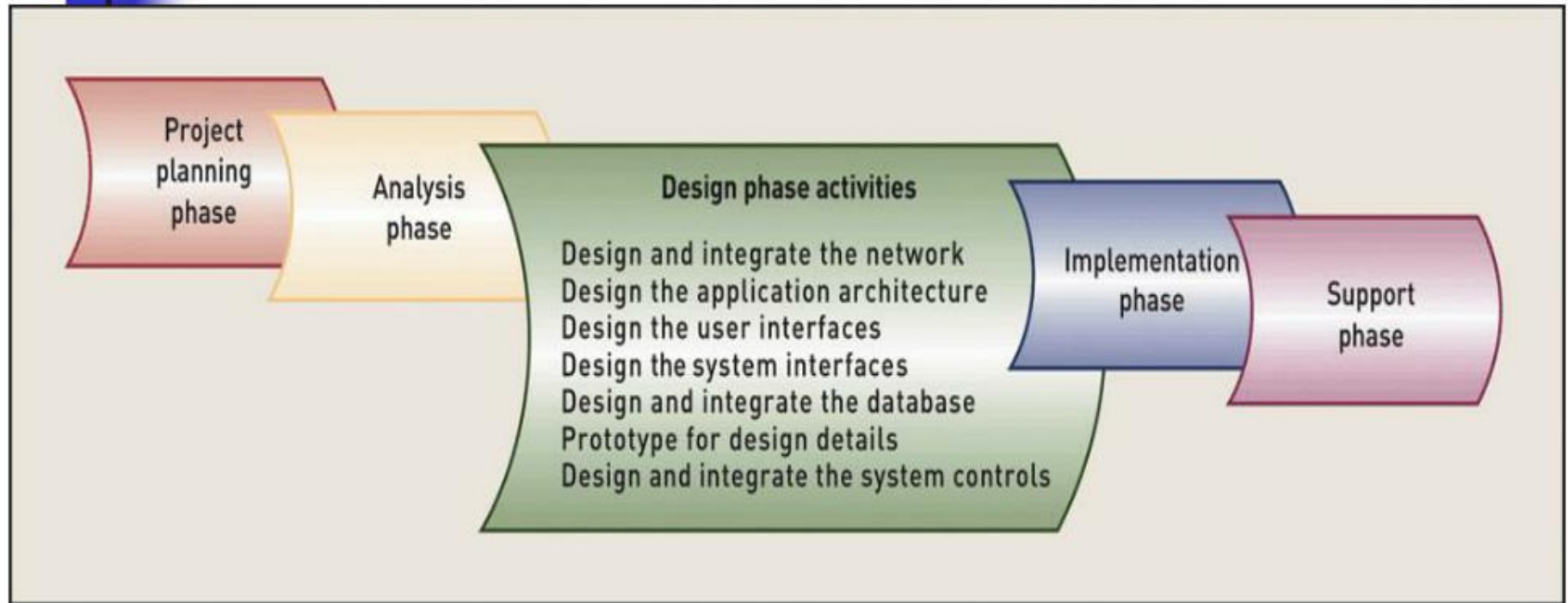


Figure 9-4

SDLC phases with design
phase activities

Design Phase Activities and Key Questions

Design phase activity	Key question
Design and integrate the network	<i>Have we specified in detail how the various parts of the system will communicate with each other throughout the organization?</i>
Design the application architecture	<i>Have we specified in detail how each system activity is actually carried out by the people and computers?</i>
Design the user interface(s)	<i>Have we specified in detail how all users will interact with the system?</i>
Design the system interface(s)	<i>Have we specified in detail how the system will work with all other systems inside and outside our organization?</i>
Design and integrate the database	<i>Have we specified in detail how and where the system will store all of the information needed by the organization?</i>
Prototype for design details	<i>Have we created prototypes to ensure all detailed design decisions have been fully understood?</i>
Design and integrate the system controls	<i>Have we specified in detail how we can be sure that the system operates correctly and the data maintained by the system are safe and secure?</i>



Database Design

Database Application Example

Microsoft Access

File Edit View Insert Format Records Tools Window Help Show Me

Type a question for help

D:\Teaching\k723\samledb\Northwind97.mdb

Northwind97 : Database (Access 97 file format)

Open Design New

Objects

- Tables
- Queries
- Forms
- Reports

Categories

- Categories
- Customer Labels Dialog
- Customer Orders
- Customer Orders Subform1
- Customer Orders Subform2
- Customer Phone List
- Orders
- Orders Subform
- Product List
- Products
- Quarterly Orders
- Quarterly Orders Subf

Main Switchboard

View Product and Order Information:

Categories Suppliers

Products Orders

Print Sales Reports

Display Database Window

Exit Microsoft Access

Categories

Category Name: Grains/Cereals

Description: Breads, crackers, pasta, and cereal

Product Name: Fil

Quantity Per Unit: 16

Product Name: Gr

Quantity Per Unit: 24

Orders

Bill To: Alfreds Futterkiste

Ship To: Alfreds Futterkiste

Obere Str. 57

Berlin 12209

Germany

Salesperson: Suyama, Michael

Ship Via: ☒ Speedy ☐ United ☐ Federal

Order ID: 10643 Order Date: 25-Sep-95 Required Date: 23-Oct-95 Shipped Date: 03-Oct-95

Product	Unit Price	Quantity	Discount	Extended Price
Spegesild	\$12.00	2	25%	\$18.00
Chartreuse verte	\$18.00	21	25%	\$283.50
Rössle Sauerkraut	\$45.60	15	25%	\$513.00
			0%	

Record: 14 of 8

Form View

Start

Student

5:00 PM

<http://www.gtaa.com/webfids/arrivals.htm>

Flight Arrivals at Toronto International Airport - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Links Customize Links Free Hotmail Windows Media Windows

Back Forward Stop Refresh Home Search Favorites Media History Mail Print Edit

Address <http://www.gtaa.com/webfids/arrivals.htm> Go

Greater Toronto Airports Authority 1280 CFYZ Listen to live radio

Flight Arrivals @ Toronto Pearson International Airport

[HOME] [DEPARTURES]

Airline	Flight	Sched Date	Sched Time	Revised Time	Arriving From	Terminal	Remarks
AIR CANADA	AC0887	19Aug	06:10	06:23	TEL AVIV	1	Arrived
AIR CANADA	AC0794	19Aug	06:15	06:24	LOS ANGELES	2	Arrived
AIR NEW ZEALAND	NZ4294	19Aug	06:15	06:24	LOS ANGELES	2	Arrived
UNITED	UA3024	19Aug	06:15	06:24	LOS ANGELES	2	Arrived
AIR CANADA	AC0156	19Aug	06:21	06:13	VANCOUVER	2	Arrived
AIR CANADA	AC0146	19Aug	06:23	06:15	CALGARY	2	Early
AIR CANADA	AC0160	19Aug	06:23	06:11	EDMONTON	2	Arrived
AIR CANADA	AC0754	19Aug	06:25	06:12	SAN FRANCISCO	2	Arrived
UNITED	UA3218	19Aug	06:25	06:12	SAN FRANCISCO	2	Arrived
Tango	AC6078	19Aug	06:34	06:42	CALGARY	3	Arrived
JETSGO	SG0100	19Aug	06:35	06:44	VANCOUVER	3	Arrived
WESTJET	WS0652	19Aug	06:35	06:47	CALGARY	3	Arrived
WESTJET	WS0656	19Aug	06:35	06:46	VANCOUVER EDMONTON	3	Arrived
AIR CANADA	AC0991	19Aug	06:40	06:40	SAO PAULO	1	Arrived
Tango	AC6074	19Aug	06:43	06:50	VANCOUVER	3	Arrived
AMERICAN	AA0456	19Aug	07:01	07:01	LOS ANGELES	3	Arrived

Local intranet

Start Student 5:06 PM

<http://www.1800flowers.com/Flowers/welcome.asp?section=1>

1-800-FLOWERS.COM - Flowers, Plants, Gourmet and Sweets, Unique Gifts and Specialty Boutiques - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Links Customize Links Free Hotmail Windows Media Windows

Back Forward Stop Refresh Home Search Favorites Media History Mail Print Edit

Address <http://www.1800flowers.com/Flowers/welcome.asp?section=1> Go

1-800-flowers.com
your trusted guide to gifting™

account login shopping basket Basket is Empty

gift reminder register now member benefits track your order

home flowers plants gourmet sweet shoppe unique gifts specialty boutiques search:

flowers

roses starting at \$29.99!

In Season

- Rose Celebration
- Summer
- National Friendship Week
- Back to School
- Best Sellers
- Same Day Gifting
- Big Summer Sale
- American Spirit

Everyday Celebrations

- Birthday
- Wedding
- Anniversary
- New Baby
- Business Gifts
- Gifts for Him

Sentiments

- Get Well
- Thank You
- Congratulations
- Love & Romance
- Just Because
- Sympathy

Department

- Arrangements
- Balloon Boutique®
- Dried Flowers
- Roses
- 1-800-THE-ROSES™
- Flower Country

featured flower products:

Summer Bouquet
The essence of summer's blooms.
\$44.99
[buy now](#)

Mille de Fleur Arrangement
Many varieties make it lovely.
\$59.99
[buy now](#)

Pink Perfection™
Feminine splendor.
ON SALE
\$47.99
[buy now](#)

Romantic Red
Stunning roses, great savings!
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\$39.99
\$34.99
[buy now](#)

Make Lemonade™
Yellow flowers with a twist!
\$39.99
[buy now](#)

One Dozen Multicolored Roses
Pure rose romance.
\$49.99 - \$99.99
[buy now](#)

Local intranet

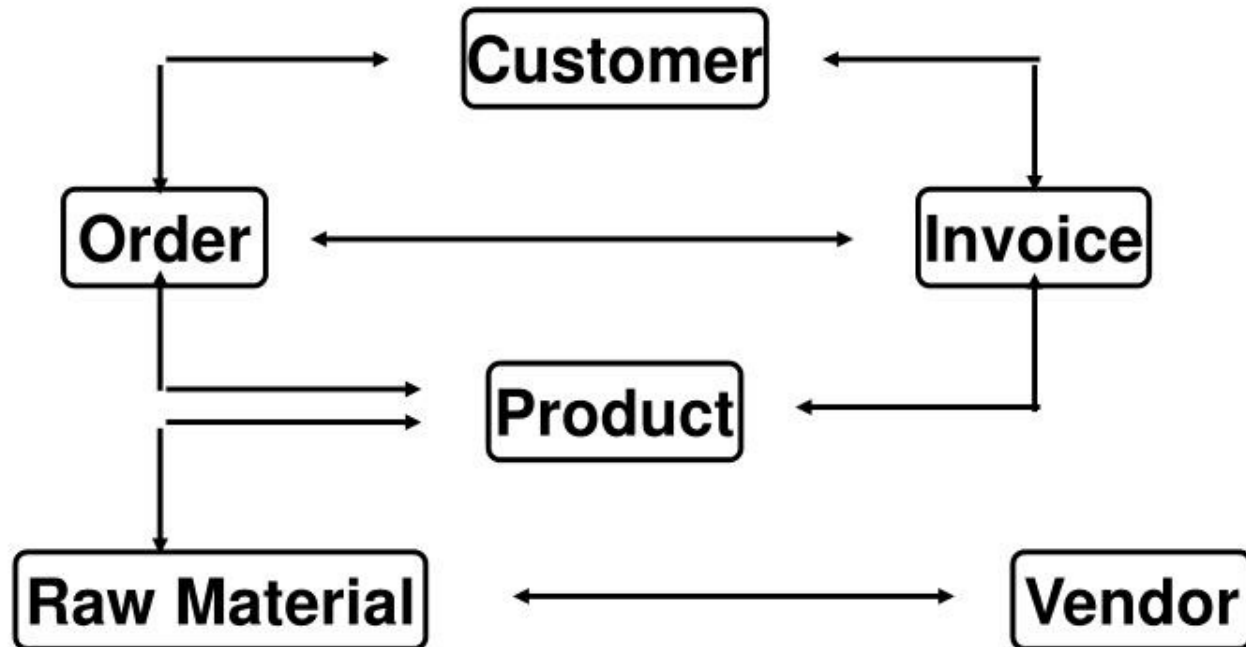
Start Student 5:16 PM



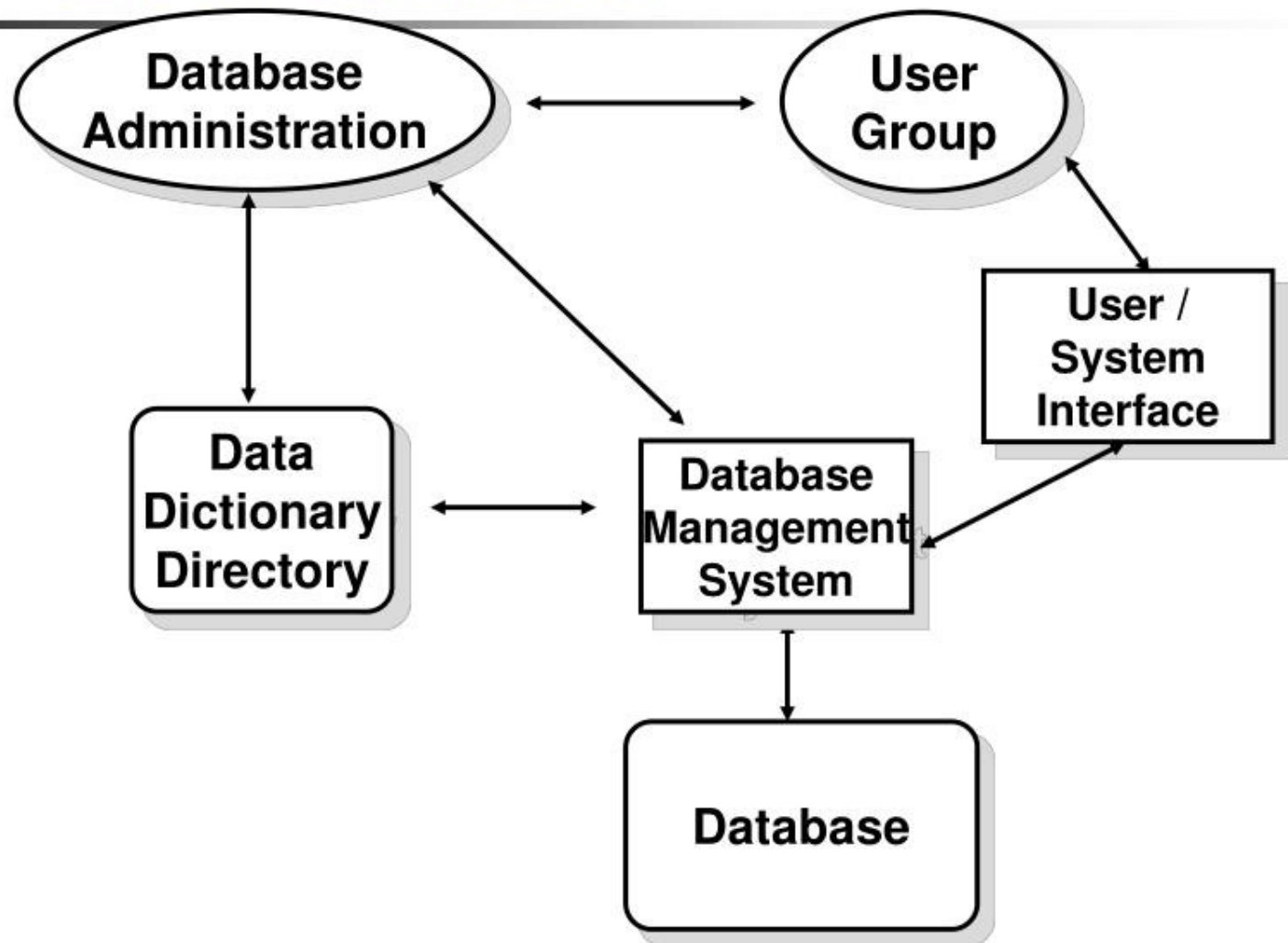
Database - A Definition

- A database is a shared collection of interrelated data, designed to meet the needs of multiple types of end users.
 - The data are stored so that they are independent of the program that use them.
 - A common and controlled approach is used in adding new data and modifying and retrieving existing data.
 - A database is not only shared by multiple users, but it is perceived differently by different users.
- Martin, 1981 -

A Database Approach



Components of a Database Environment



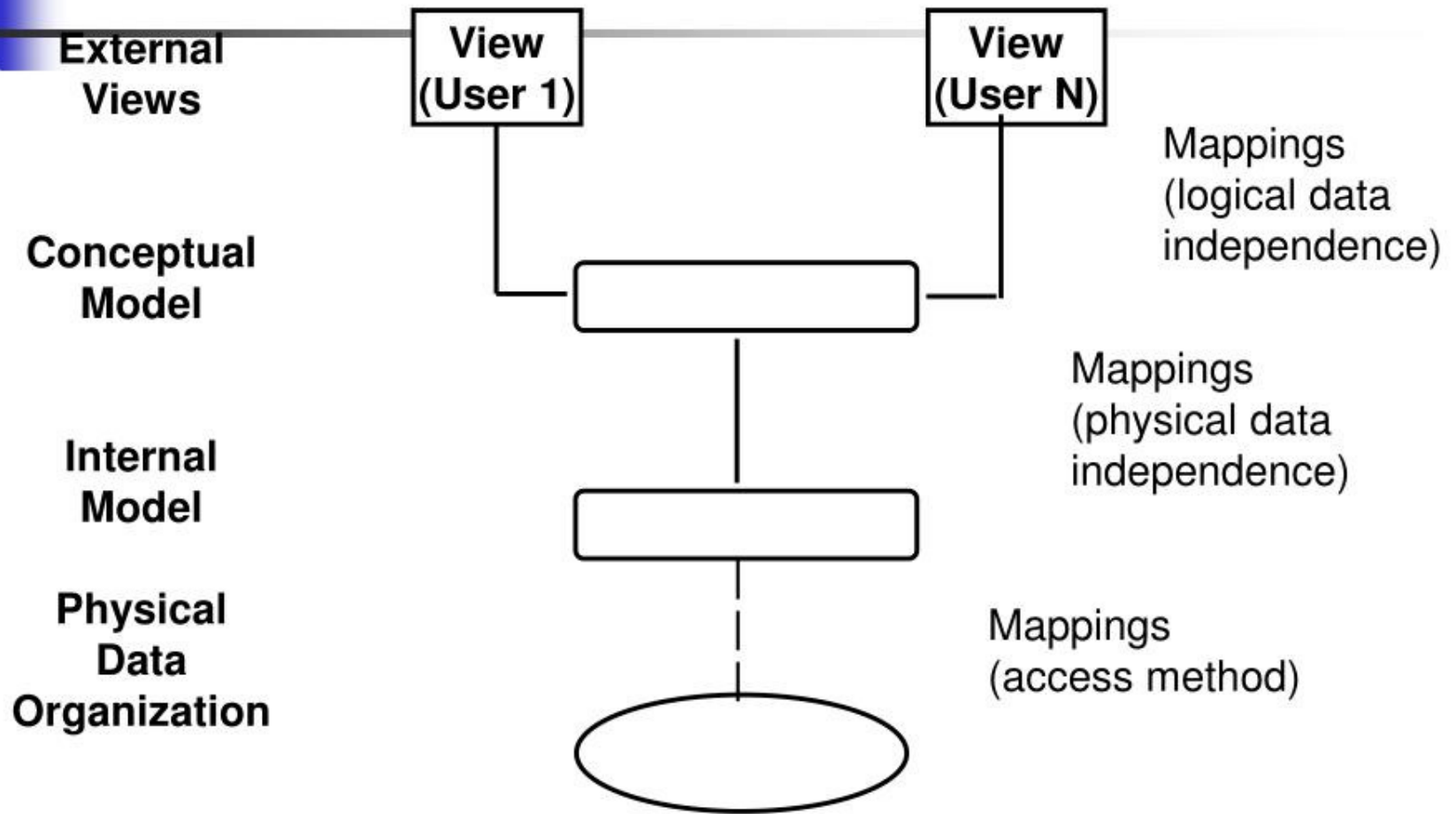


Functions of Database Management System

- Store, retrieve, and update data
- Provide integrity services to enforce database constraints
- Provide a user-accessible catalog of data descriptions
- Control concurrent processing
- Support logical transactions
- Recover from failure
- Provide security facilities
- Interface with communication control programs
- Provide utility services

Levels (or Views) of Data

ANSI/SPARC Model





ANSI / SPARC Model

- American National Standards Institute (ANSI)
- Standards Planning and Requirements Committee (SPARC)
- The conceptual model defines the entire information resource at an abstract level.
- Each user view is a subset of the conceptual model, defined and formatted according to that user's needs.
- The internal model is a definition of the physical implementation of the data base by the DBMS (Schema, Subschema)
- The physical data organization determines how data are actually organized and stored

External user views

Microsoft Access

File Edit View Insert Format Records Tools Window Help Show Me

Type a question for help

Categories

Category Name: Beverages

Description: Soft drinks, coffees, teas, beers, and ales

Picture:

Record: 1 of 8

Orders

Bill To: Alfreds Futterkiste
Obere Str. 57
Berlin 12209
Germany

Ship To: Alfreds Futterkiste
Obere Str. 57
Berlin 12209
Germany

Ship Via: ☒ Speedy ☐ United ☐ Federal

Salesperson: Suyama, Michael

Order ID: 10643 Order Date: 25-Sep-95 Required Date: 23-Oct-95 Shipped Date: 03-Oct-95

Product	Unit Price	Quantity	Discount	Extended Price
Spegesild	\$12.00	2	25%	\$18.00
Chartreuse verte	\$18.00	21	25%	\$283.50
Rössle Sauerkraut	\$45.60	15	25%	\$513.00
			0%	

Subtotal: \$814.50
Freight: \$29.46
Total: \$843.96

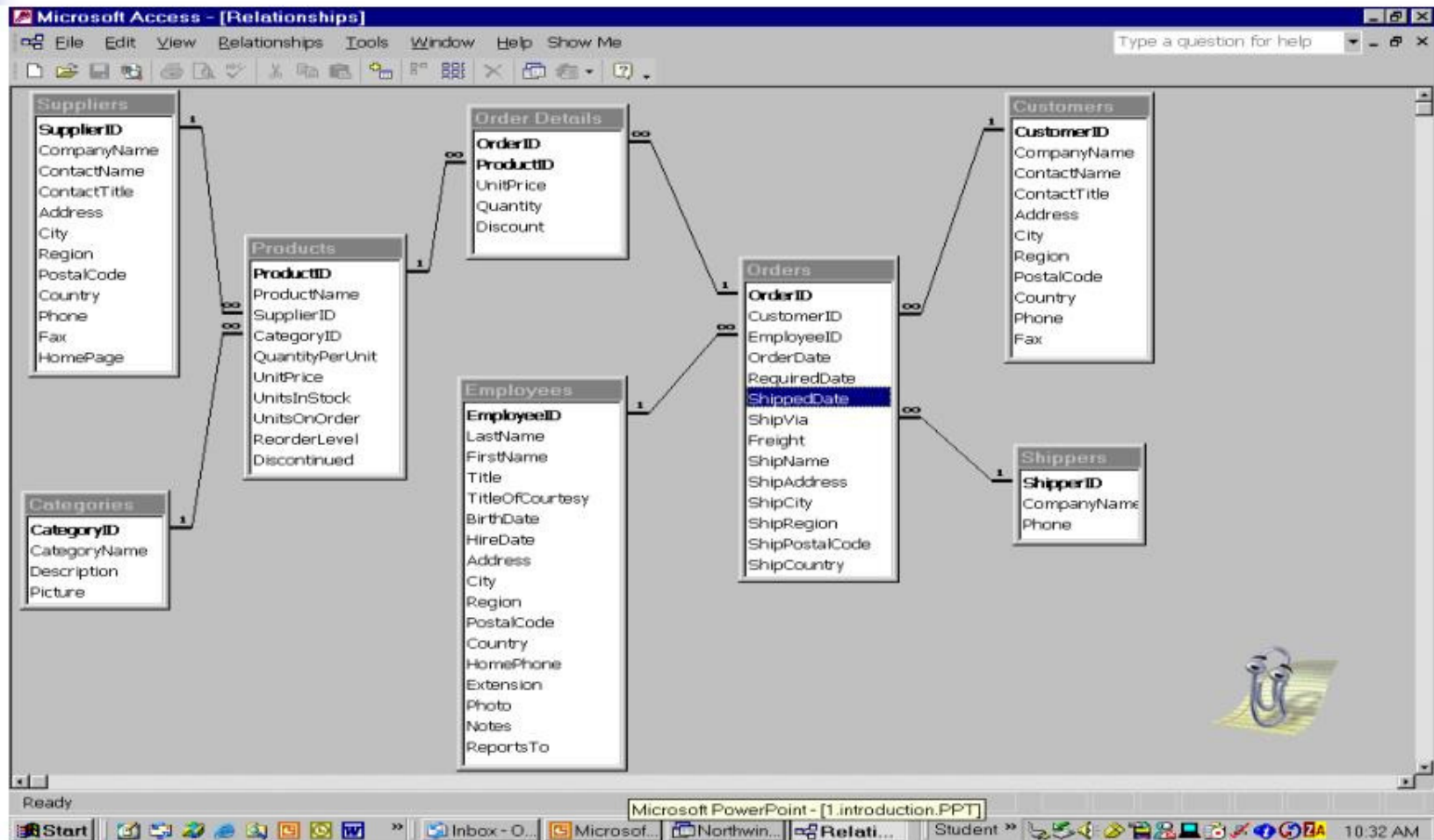
Display products of the month Print Invoice

Record: 1 of 830

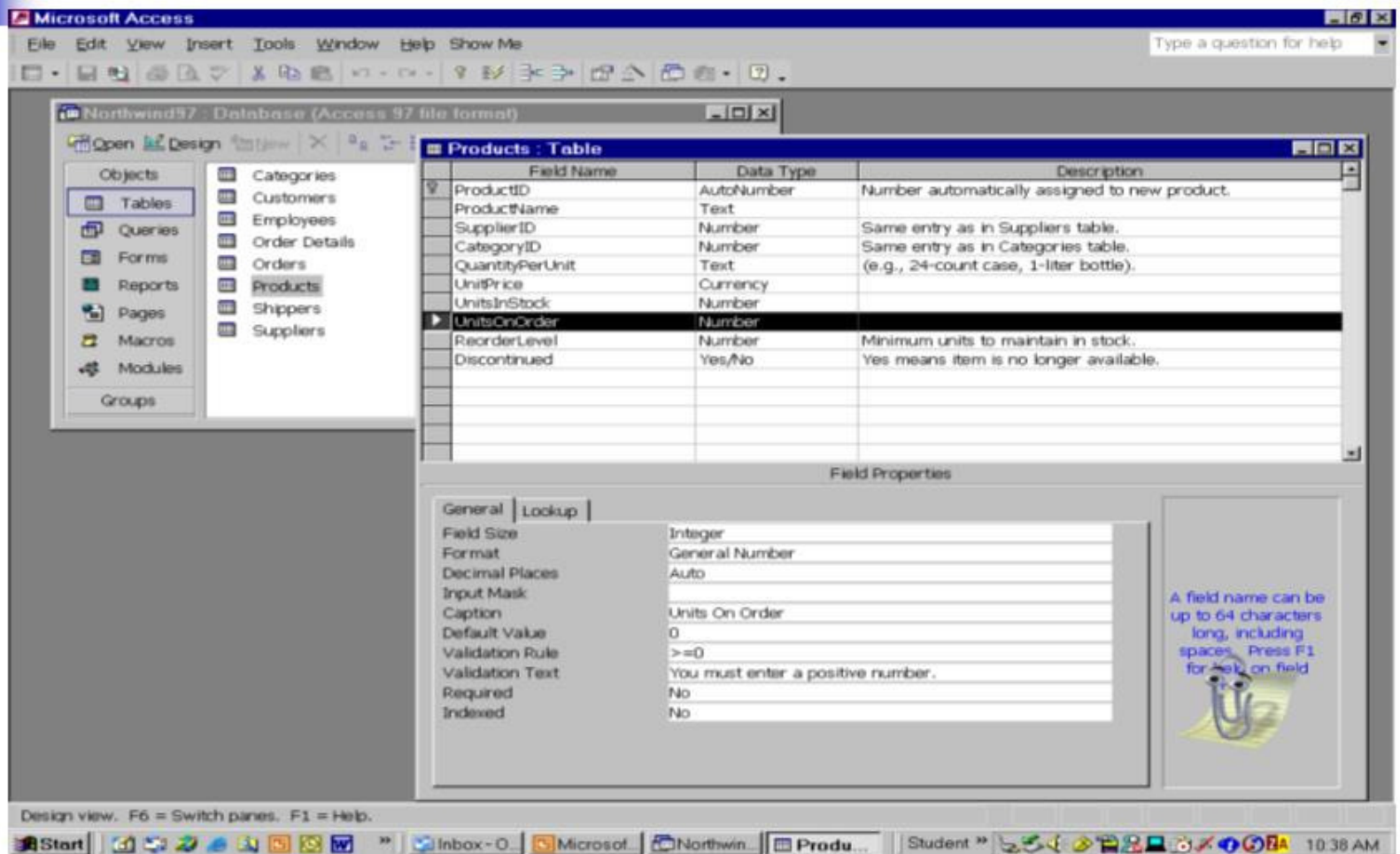
Select or type a customer's name.

Start Inb... Mic... Nor... Mai... Cat... Or... Student 10:25 AM

Conceptual Model



Internal Model



Physical Data Organization

Microsoft Access - [Products : Table]

File Edit View Insert Format Records Tools Window Help Show Me Type a question for help

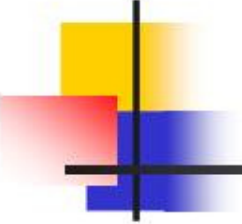
Product ID	Product Name	Supplier	Category	Quantity Per Unit	Unit Price
1	Chai	Exotic Liquids	Beverages	10 boxes x 20 bags	\$18.00
2	Chang	Exotic Liquids	Beverages	24 - 12 oz bottles	\$19.00
3	Aniseed Syrup	Exotic Liquids	Condiments	12 - 550 ml bottles	\$10.00
4	Chef Anton's Cajun Seasoning	New Orleans Cajun Delights	Condiments	48 - 6 oz jars	\$22.00
5	Chef Anton's Gumbo Mix	New Orleans Cajun Delights	Condiments	36 boxes	\$21.35
6	Grandma's Boysenberry Spread	Grandma Kelly's Homestead	Condiments	12 - 8 oz jars	\$25.00
7	Uncle Bob's Organic Dried Pears	Grandma Kelly's Homestead	Produce	12 - 1 lb pkgs.	\$30.00
8	Northwoods Cranberry Sauce	Grandma Kelly's Homestead	Condiments	12 - 12 oz jars	\$40.00
9	Mishi Kobe Niku	Tokyo Traders	Meat/Poultry	18 - 500 g pkgs.	\$97.00
10	Ikura	Tokyo Traders	Seafood	12 - 200 ml jars	\$31.00
11	Queso Cabrales	Cooperativa de Quesos 'Las Cabras	Dairy Products	1 kg pkg.	\$21.00
12	Queso Manchego La Pastora	Cooperativa de Quesos 'Las Cabras	Dairy Products	10 - 500 g pkgs.	\$38.00
13	Konbu	Mayumi's	Seafood	2 kg box	\$6.00
14	Tofu	Mayumi's	Produce	40 - 100 g pkgs.	\$23.25
15	Genen Shouyu	Mayumi's	Condiments	24 - 250 ml bottles	\$15.50
16	Pavlova	Pavlova, Ltd.	Confections	32 - 500 g boxes	\$17.45
17	Alice Mutton	Pavlova, Ltd.	Meat/Poultry	20 - 1 kg tins	\$39.00
18	Carnarvon Tigers	Pavlova, Ltd.	Seafood	16 kg pkg.	\$62.50
19	Teatime Chocolate Biscuits	Specialty Biscuits, Ltd.	Confections	10 boxes x 12 pieces	\$9.20
20	Sir Rodney's Marmalade	Specialty Biscuits, Ltd.	Confections	30 gift boxes	\$81.00
21	Sir Rodney's Scones	Specialty Biscuits, Ltd.	Confections	24 pkgs. x 4 pieces	\$10.00
22	Gustaf's Knäckebröd	PB Knäckebröd AB	Grains/Cereals	24 - 500 g pkgs.	\$21.00
23	Tunnbröd	PB Knäckebröd AB	Grains/Cereals	12 - 250 g pkgs.	\$9.00
24	Guaraná Fantástica	Refrescos Americanas LTDA	Beverages	12 - 355 ml cans	\$4.50
25	NuNuCa Nuß-Nougat-Creme	Heli Süßwaren GmbH & Co. KG	Confections	20 - 450 g glasser	\$14.00
26	Gumbär Gummibärchen	Heli Süßwaren GmbH & Co. KG	Confections	100 - 250 g bags	\$31.23
27	Schoggi Schokolade	Heli Süßwaren GmbH & Co. KG	Confections	100 - 100 g pieces	\$43.90
28	Rössle Sauerkraut	Putzer Lebensmittelgroßmärkte AG	Produce	25 - 875 g cans	\$15.60

Record: 1 of 28

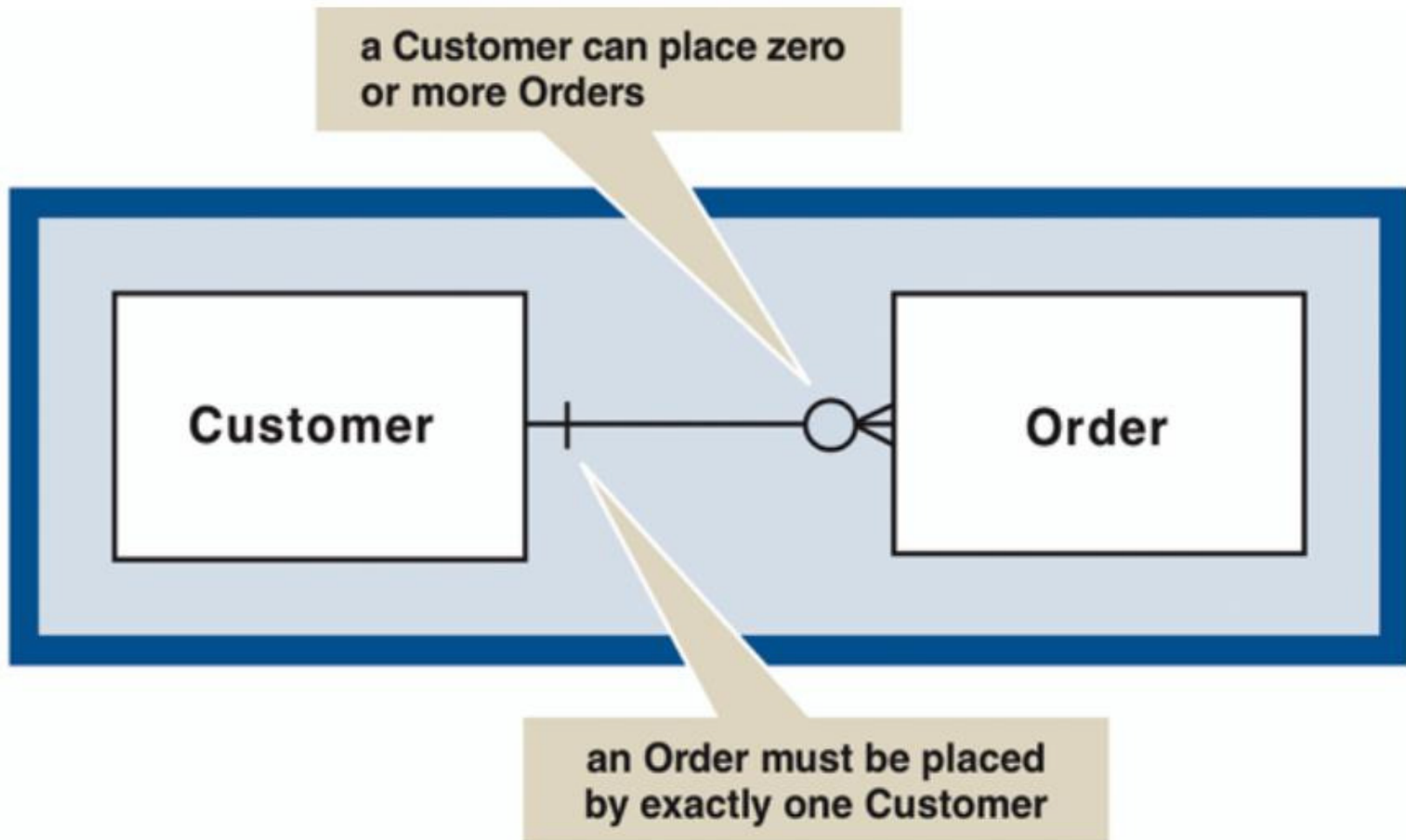
Number automatically assigned to new product.

Start | Inboxes - O... | Microsoft... | Northwin... | Produ... | Student 10:40 AM

Entity-Relationship Data Model

- 
- Entity
 - A thing, event, or person in the organization's environment about which someone wants to collect data
 - Attribute
 - Characteristic or property of an entity
 - Identifier: used to uniquely identify an entity
 - Relationship
 - Association between entities
 - Can be one to one, one to many or many to many

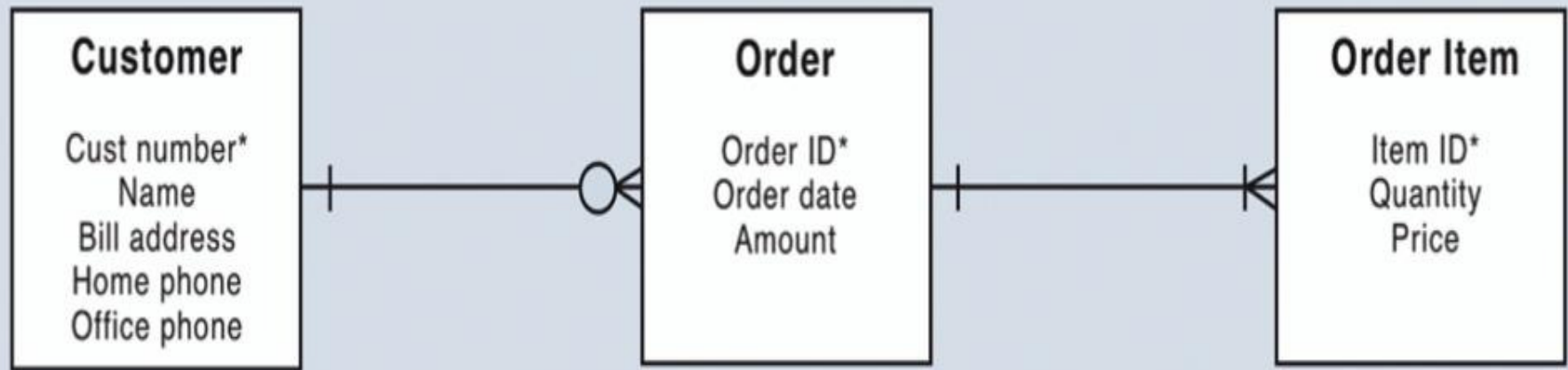
The Entity-Relationship Diagram (ERD)



Expanded ERD with Attributes Shown

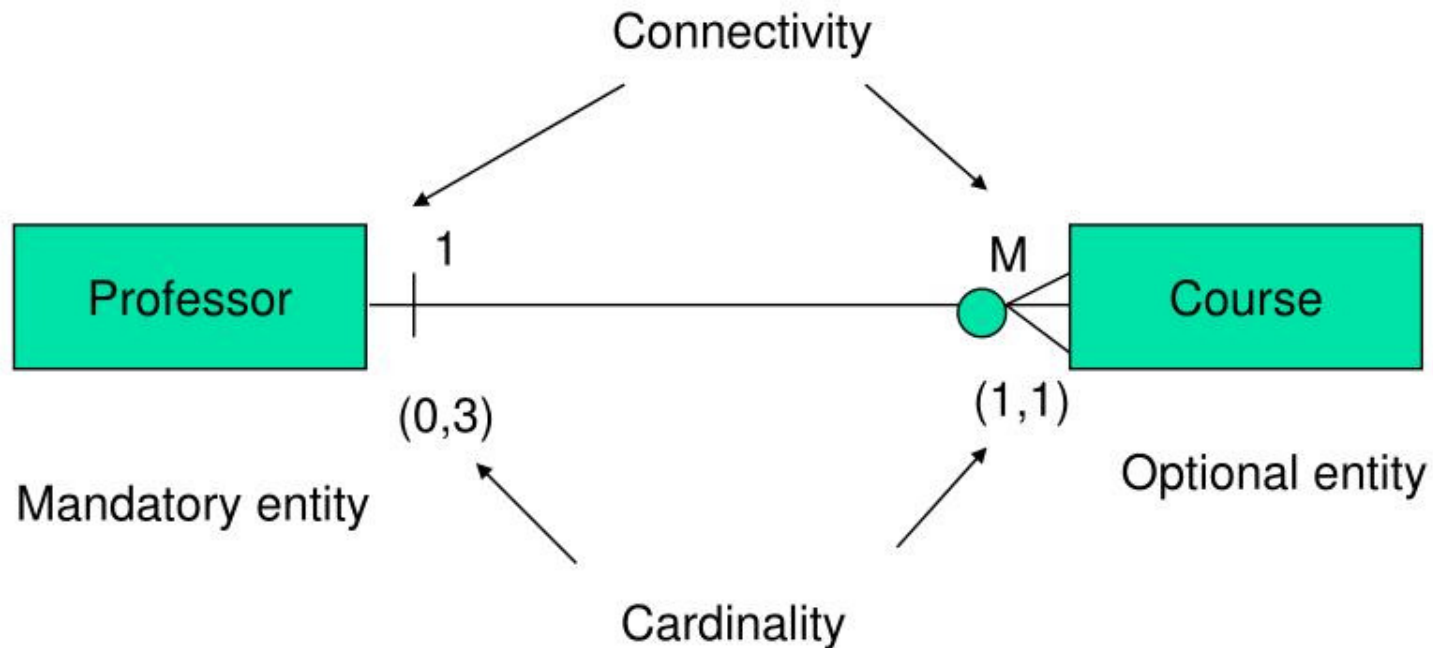
Figure 5-25

An expanded ERD with attributes shown



*Indicates the identifier or key

Connectivity and Cardinality

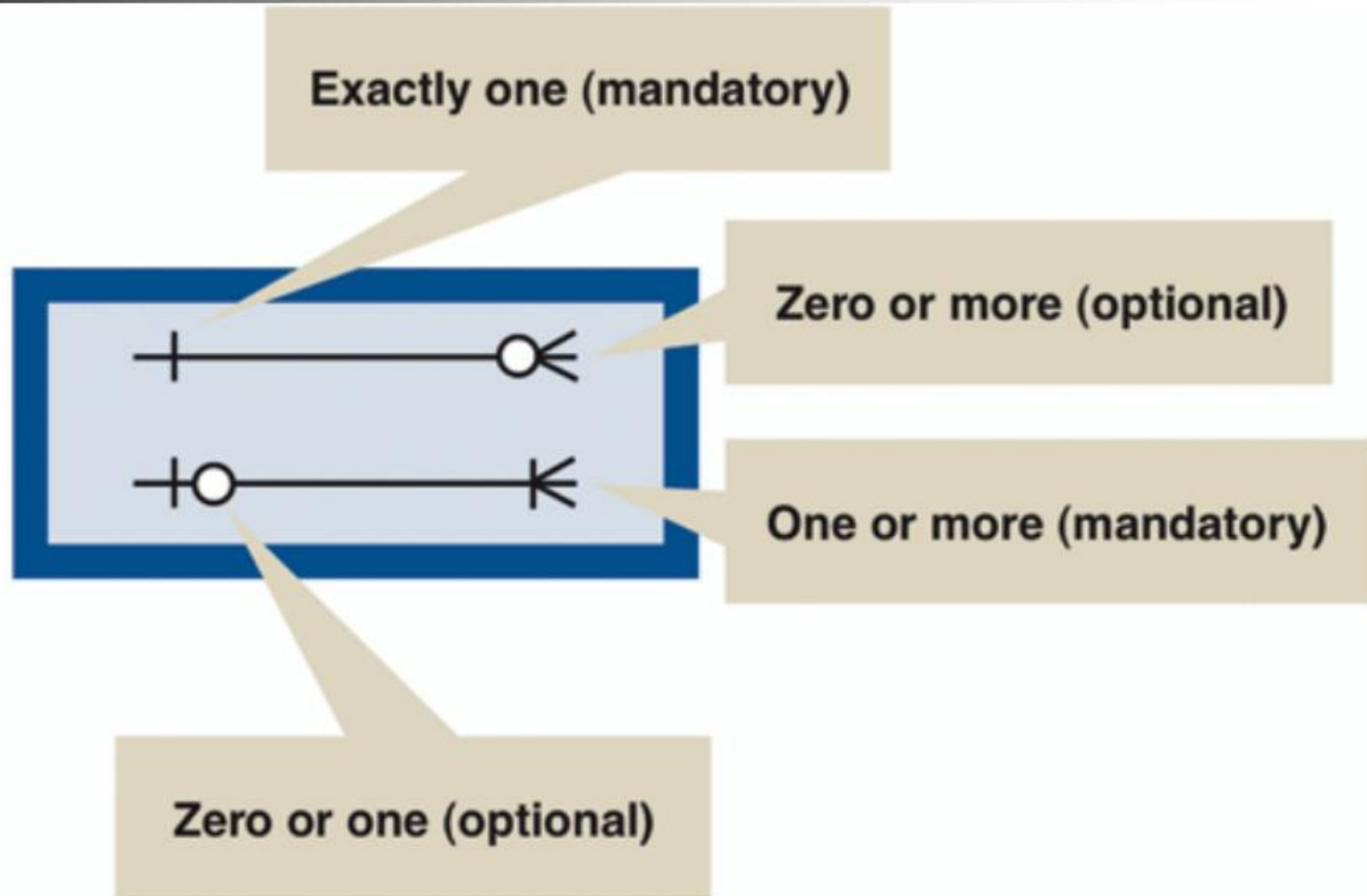




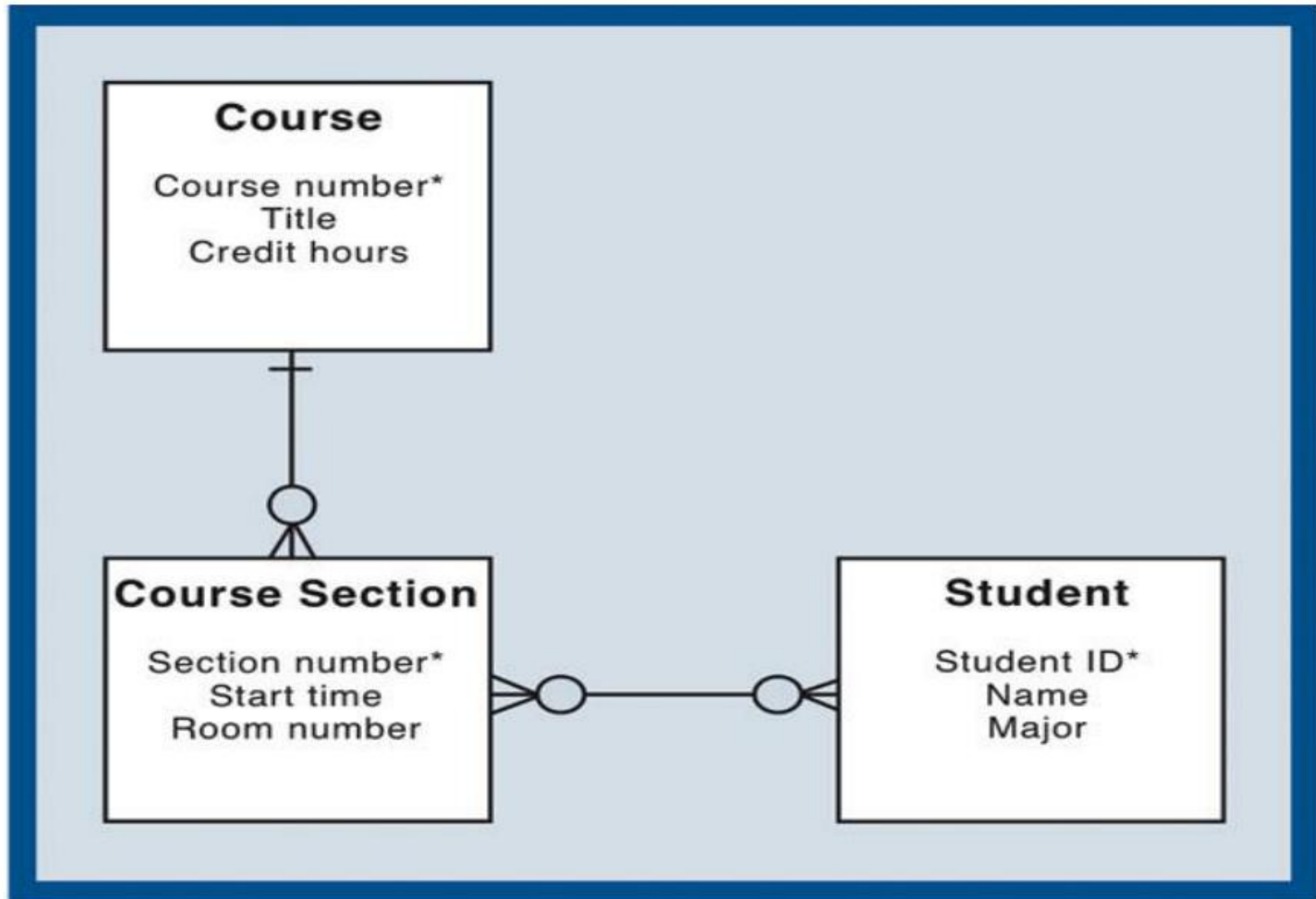
Connectivity and Cardinality

- The term connectivity is used to describe the relationship classification (e.g., one-to-one, one-to-many, and many-to-many).
- Cardinality expresses the specific number of entity occurrences associated with one occurrence of the related entity.
- Relational Participation: Optional, Mandatory
- The actual number of associated entities is usually a function of an organization's policy.

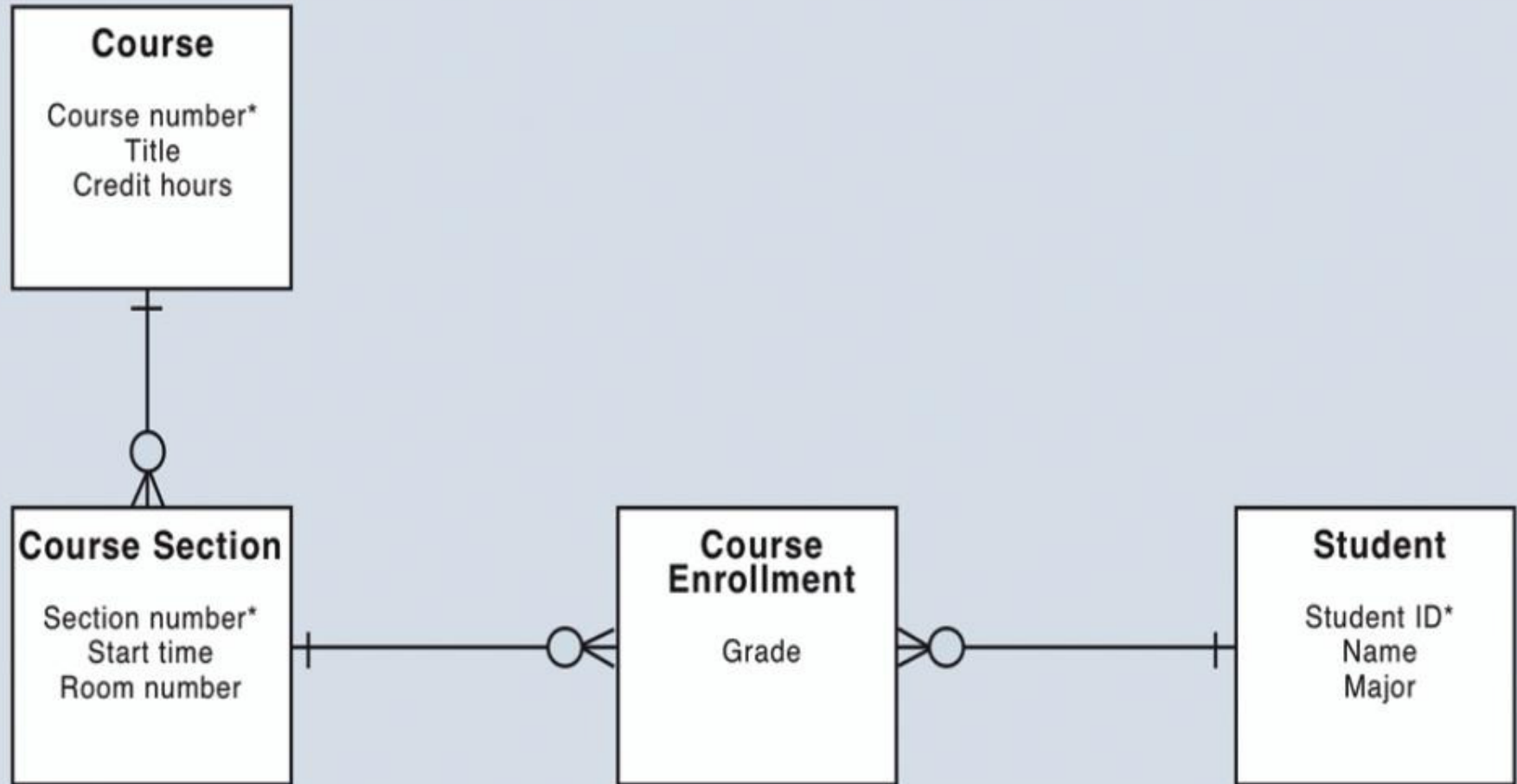
Cardinality Symbols of Relationships for ERD



ERD with Many-to-Many Relationship



Many-to-Many Relationship Converted to Associative Entity to Store Grade Attribute





Process of developing a E-R Model

- Understand business rules
- Identify major entities
- Identify relationships
- Identify attributes

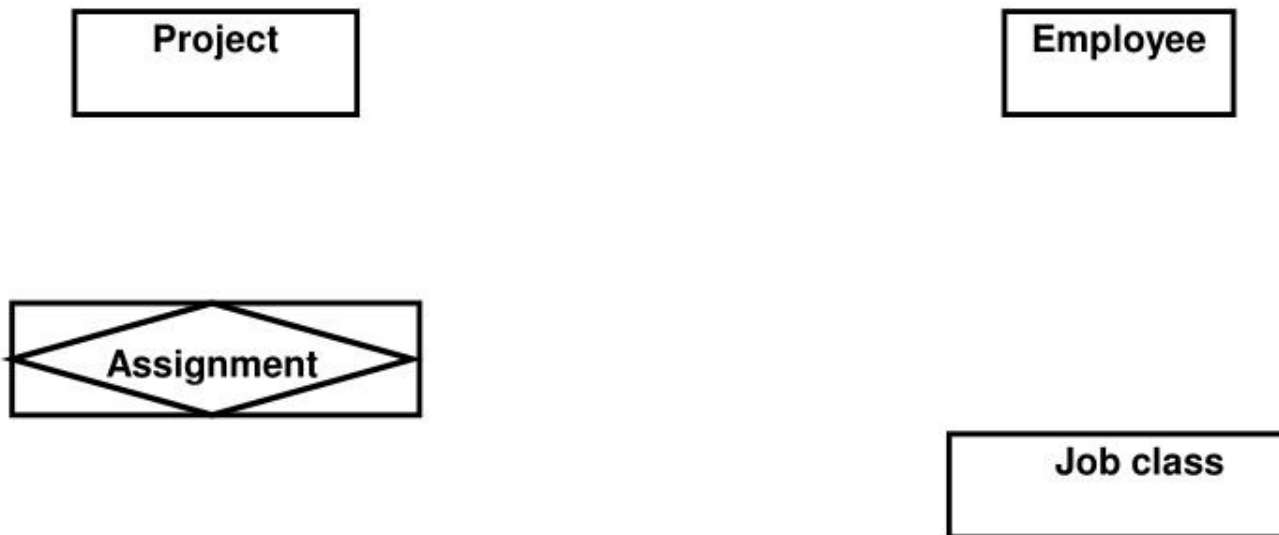


Example: A Construction Company

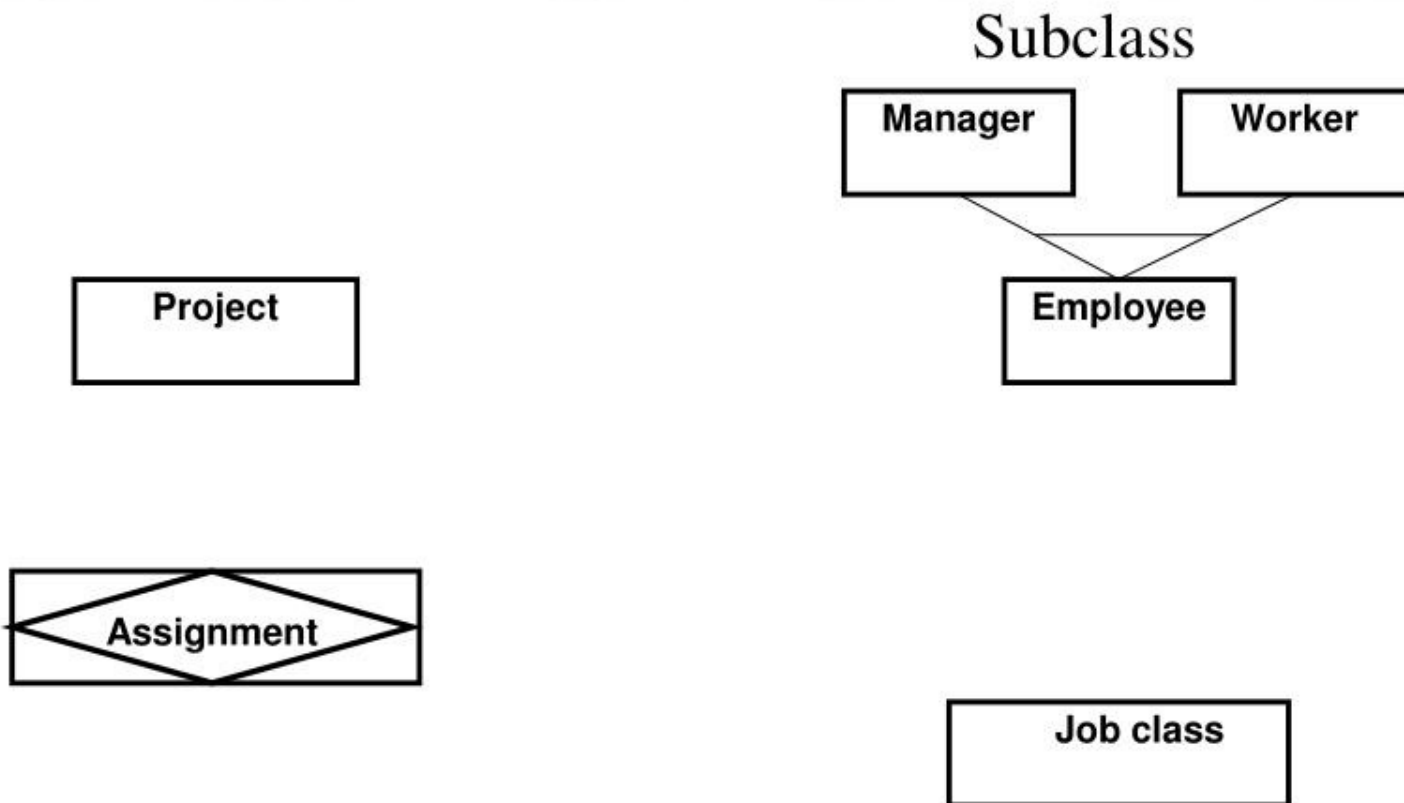
- A manager may manage many projects.
- Each project requires the services of many employees.
- An employee may be assigned to several different projects.
- Some employees are not assigned to a project and perform duties not specifically related to a project. Some employees are part of a labor pool, to be shared by all project teams.
- Each employee has a (single) primary job classification. This job classification determines the hourly billing rate.
- Many employees can have the same job classification.



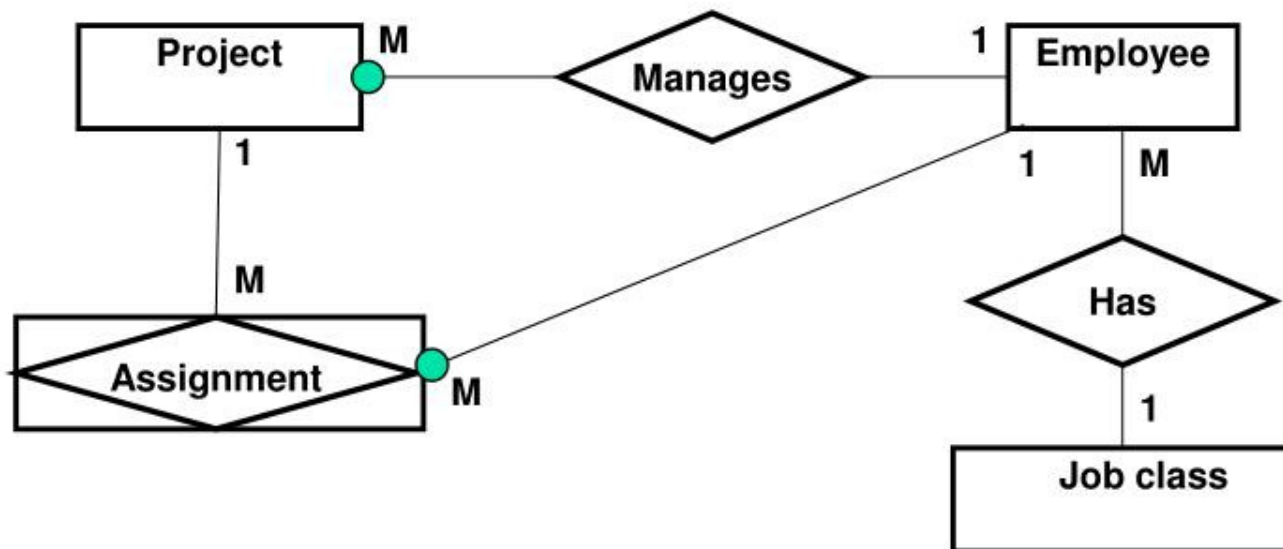
A E-R Model for A Construction Company



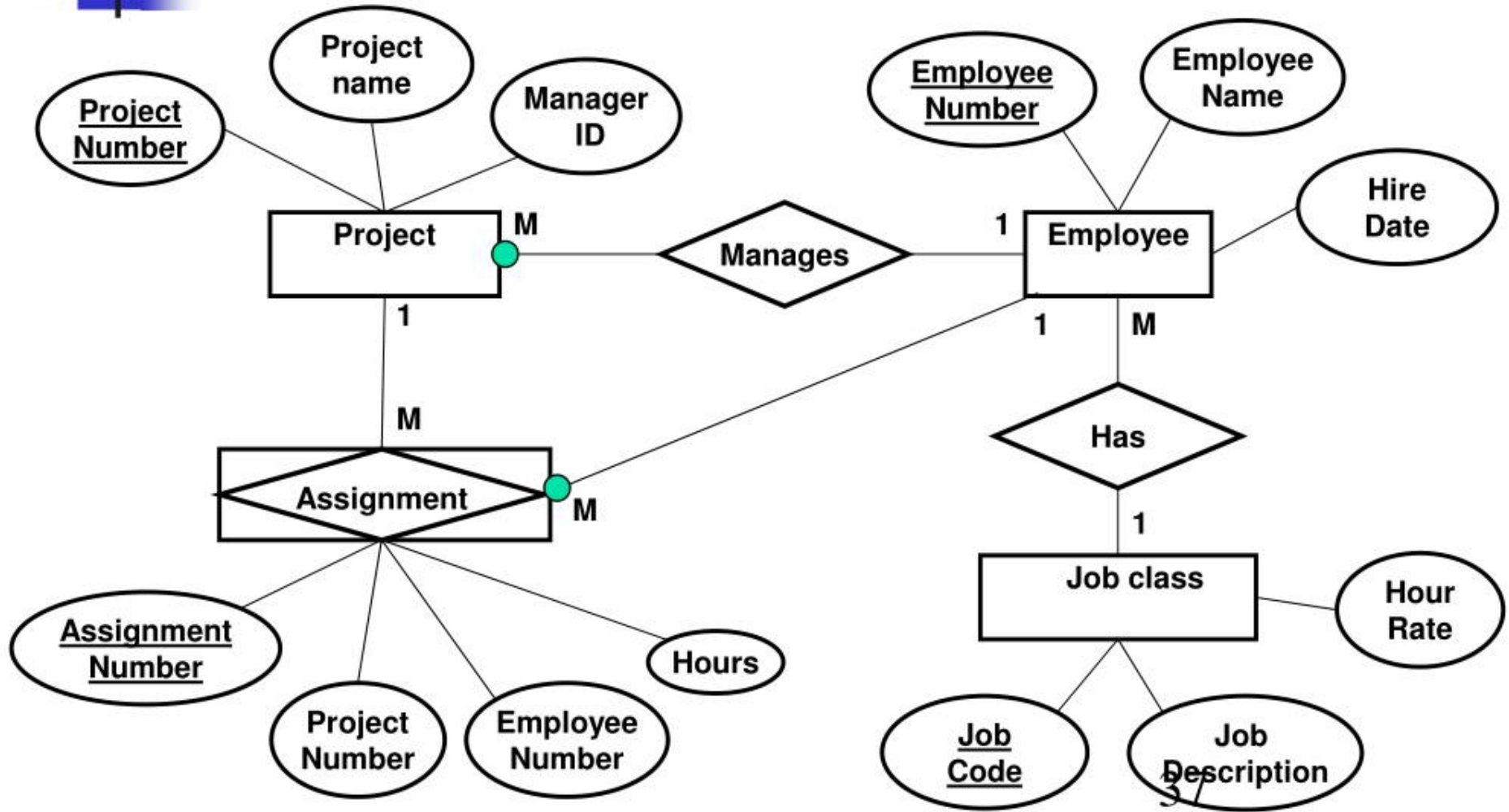
A E-R Model for A Construction Company



A E-R Model for A Construction Company



A E-R Model for A Construction Company





Relational Data Model

- A relational data model organizes data as a set of relations, or two-dimensional tables.
- A relation is viewed as a two-dimensional table, with following properties:
 - Each column contains values about the same attribute, and each table cell must be simple
 - Each column has a distinct name (attribute name), and the order of columns is immaterial
 - Each row is distinct, duplicate rows are not allowed
 - The sequence of the rows is immaterial



Sample Relational tables

STAFF table

ID	NAME	DEPT	JOB	YEARS	SALARY	COMM
10	SANDERS	D20	MGR	7	18357.50	-
20	PERNAL	D20	SALES	8	18171.25	612.45
30	MARENGHI	D38	MGR	5	17506.75	-
40	O'BRIEN	D20	SALES	6	18006.00	846.55
50	HANES	D15	MGR	10	20659.80	-
80	JAMES	D20	CLERK	-	13540.60	-

ORG table

DEPTNUMB	DEPTNAME	MANAGER	DIVISION	LOCATION
D10	HEAD OFFICE	160	CORPORATE	NEW YORK
D15	NEW ENGLAND	50	EASTERN	BOSTON
D20	MID ATLANTIC	10	EASTERN	WASHINGTON
D38	SOUTH ATLANTIC	30	EASTERN	ATLANTA



Terminology in a Relation

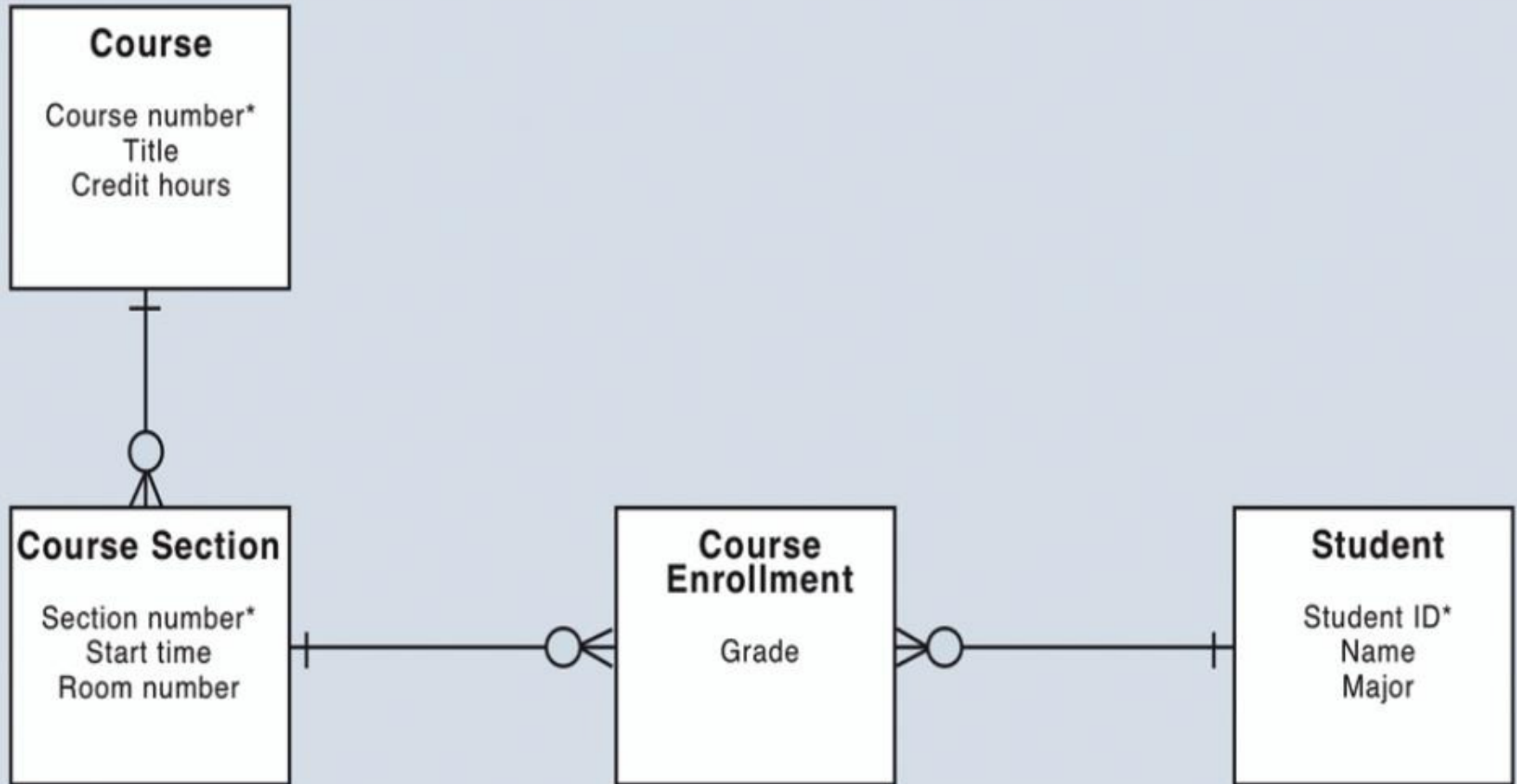
- Tuple - a row or record
- Column - values of an attribute
- Domain - a set of possible values for an attribute



Terminology in a Relation

- Key
 - primary key (unique ID)
 - Concatenated key - use two or more attributes to identify a record (e.g.. Student ID & Course ID to identify a Grade record)
- Foreign key (cross reference key)
 - a foreign key is a non-key attribute in one relation that also appears as a primary key in another relation

A E-R Diagram for Student Registration





Covert E-R Model to Relational Tables

- Create one table for each entity with key and attributes
- Introduce foreign key into the “many” side to represent one-to-many relationship
- Create new tables to represent many-to-many relationships through the corresponding foreign keys



A Relational Model For Student Registration System

Course Table

<u>Course ID</u>	Title	Credit hours
------------------	-------	--------------

Course Section Table

<u>Course ID</u>	<u>Section Number</u>	Start Time	Room Number
------------------	-----------------------	------------	-------------

Student Table

<u>Student ID</u>	Student Name	Major
-------------------	--------------	-------

Enrollment Table

<u>Course ID</u>	<u>Section Number</u>	<u>Student ID</u>	Grade
------------------	-----------------------	-------------------	-------



User Interface Design



User versus System Interface

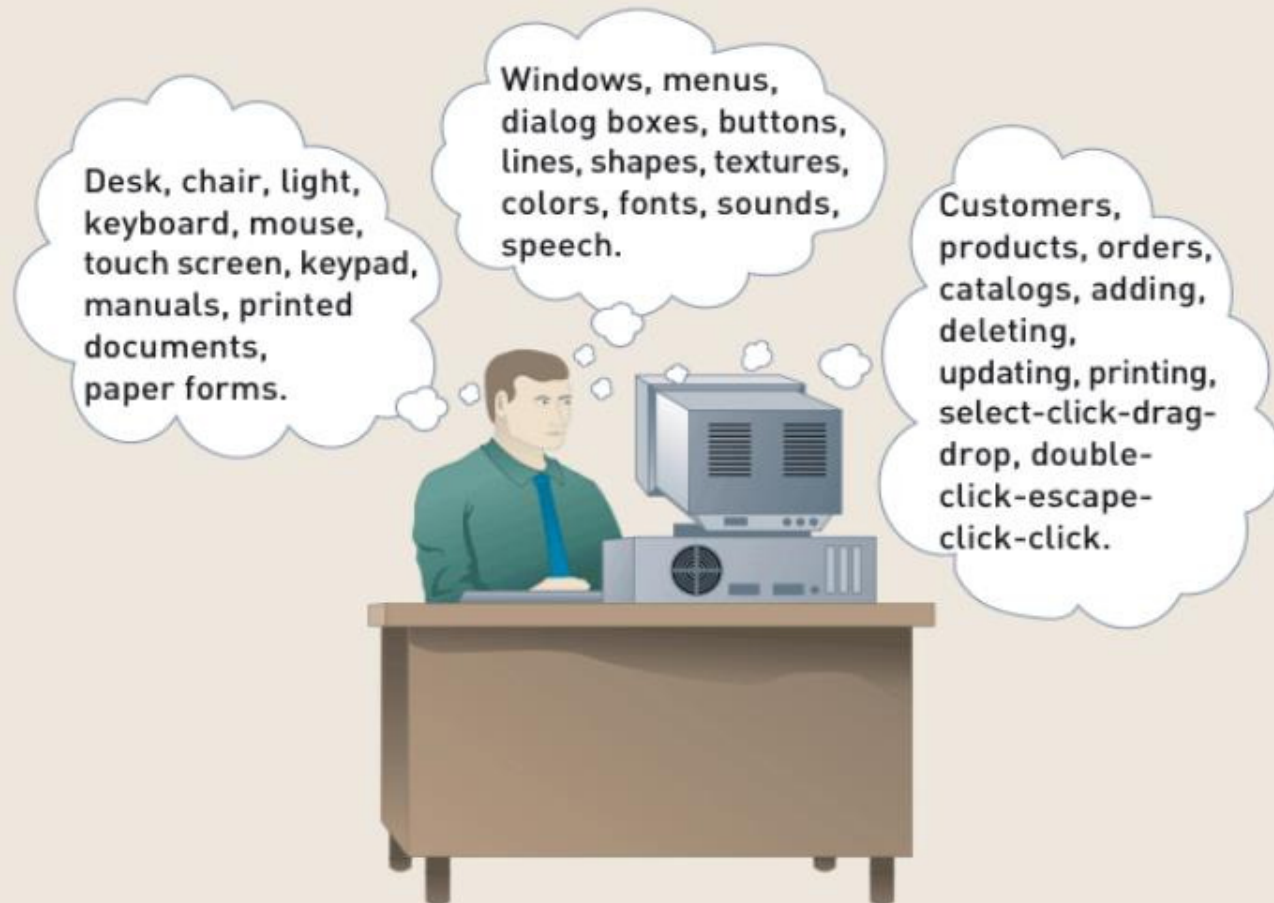
- **System interfaces** – I/O requiring minimal human interaction
- **User interfaces**
 - I/O requiring human interaction
 - User interface is everything end user comes into contact with while using the system
 - To the user, the interface *is* the system
- Analyst designs system interfaces separate from user interfaces
- Requires different expertise and technology



Understanding the User Interface

- Physical aspects of the user interface
 - Devices touched by user, computer terminals, keyboard, mouse, PDAs, cell phones,...
- Perceptual aspects of the user interface
 - Everything else user sees, hears, or touches such as screen objects, menus, and buttons
- Conceptual aspects of the user interface
 - What user knows about system and logical function of system

Aspects of the User Interface





User-Centered Design

- Focus early on the users and their work by focusing on their requirements
- Usability - system is easy to learn and use
- Iterative development keeps focus on user
 - Continually return to user requirements and evaluate system after each iteration
- Human-computer interaction (HCI)
 - Study of end users and interaction with computers
- Human factors engineering (ergonomics)



Guidelines for Designing User Interfaces

■ Visibility

- All controls should be visible
- Provide immediate feedback to indicate control is responding

■ Affordance

- Appearance of control should suggest its functionality – purpose for which it is used
- System developers should use published **interface design standards** and guidelines



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- Find co-workers, neighbors and friends for carpool
- Use for daily commute, recurring trips, long distance trips and events
- Plan ahead or use on-demand
- Track your CO2 savings through carpool
- Track your monetary savings through carpool
- Works throughout USA, Canada, Europe and Australia
- It's completely FREE

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SIGN UP



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SEARCH



CONNECT



Sharing the ride will add up!

- Carpool can result in substantial monetary savings for you. Share gas/fuel costs. Share toll costs. Share parking costs.

What is iCarpool?

iCarpool is an online service which helps you find carpool partners. Unlike a rideshare





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Is your privacy important to you? We think so. We require you to register with us to use iCarpool because our privacy controls do not allow anonymous users to acquire any member information through the website.

*** Required Fields****Provide the following information***First Name: *Last Name: *Age:

Your age will not be revealed to other members. Only an age range will be shown.

☐ Male*Sex: ☐ Female☒ Don't want to specify*Country: *State/Province: *Language: English*Phone Number: **Provide the account information***Your E-mail Address: *Confirm E-mail:

If you have an email address issued by your organization, we recommend you to provide that.

Your email address will be verified.



Helping you reduce your Carbon Footprint

[log in](#) [blog](#) [how it works](#)





Drivers and Passengers connecting online

Saving money and the environment at the same time!

Save PickupPal in Ontario!



Sign up to find someone to drive you somewhere

Need a Ride?



Sign up to find someone to share your car and reduce your eco-footprint

Give a Ride?

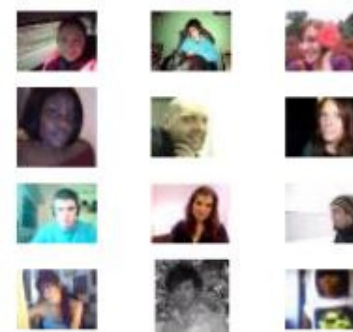
Returning pals

Email:

Password:

[log in >](#)

PickupPals — 12 Oct 2008



The friendly faces of PickupPal's newest members.

Potential emissions savings for PickupPal Eco-Rideshare



[log in](#) [blog](#) [how it works](#)



Welcome!

New passenger registration

Personal info

First name:

Last name:

Email address:

Confirm email address:

☐ I certify that I am over the age of majority in my jurisdiction

[sign up >](#)



Contact drivers in your area who can drive you right now!

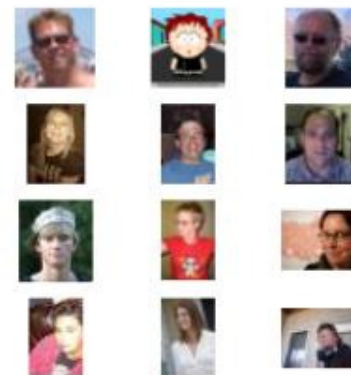
Choose who drives you and for how much.

Choose when and how often.

Control the entire process.

It's fast and **free** to join!

PickupPals — 12 Oct 2008



The friendly faces of PickupPal's newest members.



[log in](#) [blog](#) [how it works](#)

Welcome! New driver registration

Personal info

First name:

Last name:

Email address:

Confirm email address:

☐ I certify that I am over the age of majority
in my jurisdiction

[sign up >](#)



Find people in your area who need a drive
now!

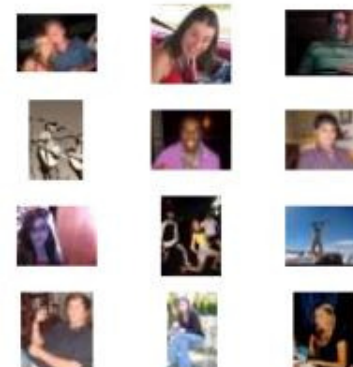
Choose your passenger or package.

Choose when and how often you drive.

Decide how much you want to be paid.

It's fast and **free** to join!

PickupPals — 12 Oct 2008



The friendly faces of PickupPal's
newest members.



Ten Good Deeds in Web Design

- Place organization's name and logo on every page and link to the homepage
- Provide a search function
- Use straightforward headlines and page titles so it is clear what page contains
- Structure page to help readers scan it
- Use hypertext to organize information into separate pages

<http://www.statefarm.ca/>

State Farm Canada® - Microsoft Internet Explorer

http://www.statefarm.ca/ Google

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State Farm Canada®

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Access your account Register Forgot Password?

Customer ID: Password: Login

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Retrieve a Saved Quote

Payment Options

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Customer service
Other Payment Options

Claims Centre

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Enter your postal code:

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What to do after an accident
How the claim process works

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State Farm Representatives are trained to help you find a comfort level with investing.

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We offer coverage tailored to your business needs. We offer fidelity bonds and surety bonds. We also insure business autos and home based businesses. [Learn more.](#)

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We offer a broad spectrum of Life Insurance and Annuities products to fit most situations. [Learn more.](#)

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Search

Claims Centre



Locate a claim office by postal code

[Go](#) [Advanced Search](#)

Report A Claim Online

Auto and Property

Select a province [Go](#)

If you have reported a claim and would like additional information regarding weather related damage, visit our online [Catastrophe](#) center.

Claims Centre

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[Auto Claims](#)

[Secure Your Car](#)

[Property Claims](#)

[Property Claims Settlement Process](#)

[Additional Suggestions](#)

Repairs

[State Farm's Auto Damage Service Programs](#)

[Find a nearby auto repair facility](#)

[Windshield Repairs](#)

[Why Competitive Bids are Used](#)

[State Farm's replacement parts promises](#)

[Home Repairs](#)

Related Links

[Disaster Preparedness](#)

[After an Accident](#)

[Loss at Home](#)

[Design Your Loss Prevention Program](#)



Ten Good Deeds in Web Design (Continued)

- Use product photos (preferably thumbnails), but avoid cluttered and bloated pages that load slowly
- Use relevance-enhanced image reduction; zoom in on needed detail
- Use link titles to provide users with a preview of where link will take them
- Ensure that pages are accessible by users with disabilities
- Do the same thing as everybody else because users come to expect certain features



Eight Golden Rules for Interactive Interface Design

- Strive for consistency
- Enable frequent users to use shortcuts
- Offer informative feedback
- Design dialogs to yield closure
- Offer simple error handling
- Permit easy reversal of actions
- Support internal locus of control
- Reduce short-term memory load



User Interface Design Principles

- The structure principle
- The simplicity principle
- The visibility principle
- The feedback principle
- The tolerance principle
- The reuse principle
- <http://www.ambysoft.com/essays/userInterfaceDesign.html>

Sample Web site Structure Chart

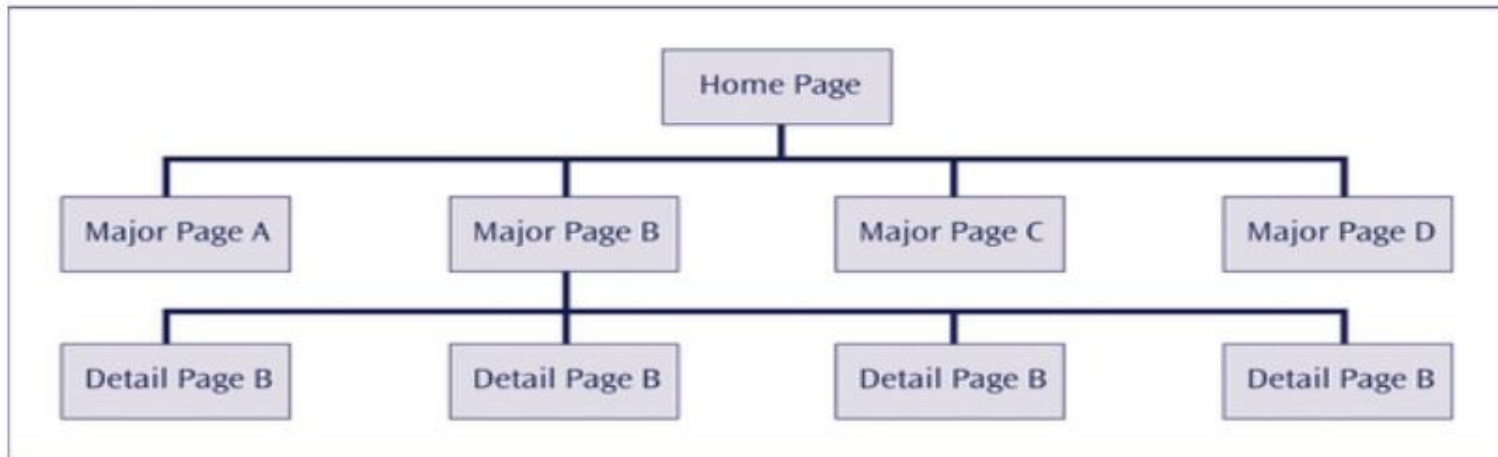


Figure 7-3
Sample Web
site structure
flow chart with
three levels



Documenting Dialog Designs

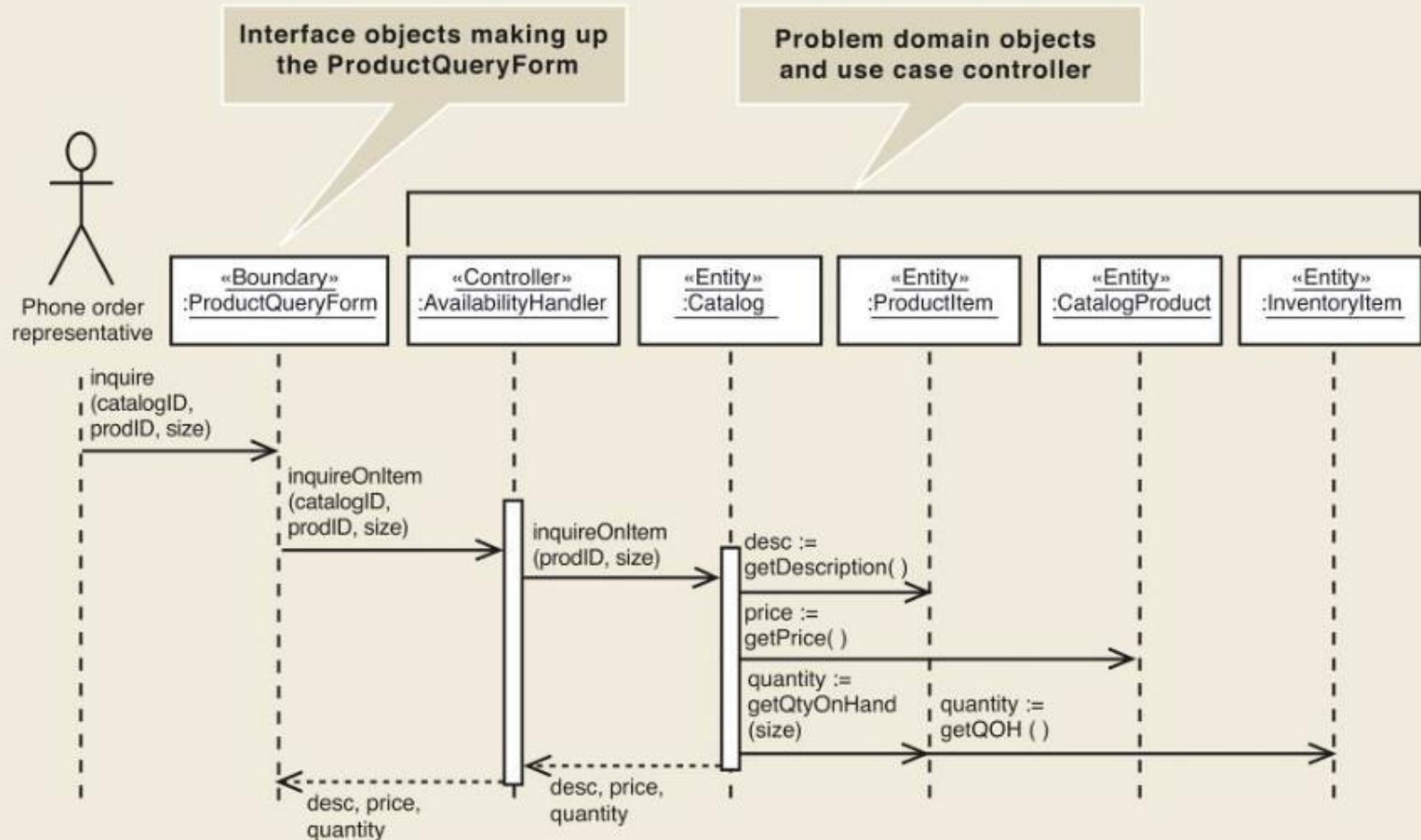
- Done simultaneously with other system activities
- Based on inputs and outputs requiring user interaction
- Used to define menu hierarchy
 - Allows user to navigate to each dialog
 - Provides overall system structure
- Storyboards, prototypes, and UML diagrams

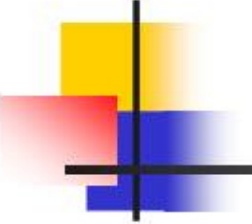


Dialogs and Storyboards

- Many methods exist for documenting dialogs
 - Written descriptions following flow of activities like in use case description
 - Narratives
 - Sketches of screens
 - **Storyboarding** – showing sequence of sketches of display screen during a dialog

Sequence Diagram for the *Look Up Item Availability* dialog





Analyzing the Web Site's Structure

- When designing the structure of a Web site, it is a good idea to create a flow chart that diagrams the Web site's navigational structure.
- A good way to do this is to use a technique called storyboarding.



What is Storyboard

- **Storyboards** are graphic organizers such as a series of illustrations or images displayed in sequence for the purpose of previsualizing a motion graphic or interactive media sequence, including website interactivity.

<http://www.thestoryboardartist.com/bullfight.html>

Josh Sheppard's Storyboard site: TV Storyboard Sample Page: Bullfight:

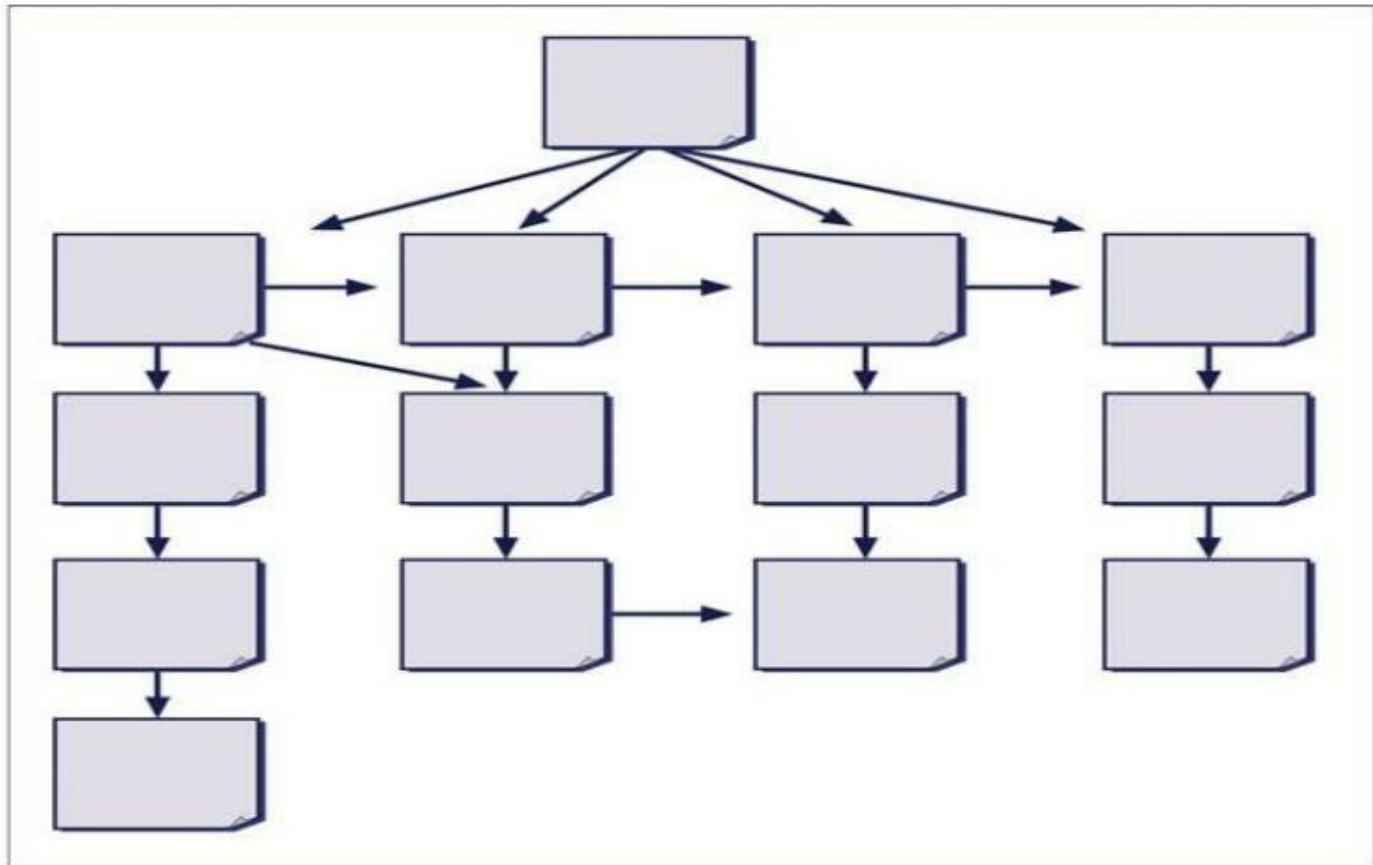
This spot was very fun to work on. It was for Tiger beer in Japan.

Click [here](#) to jump to the next TV Sample.



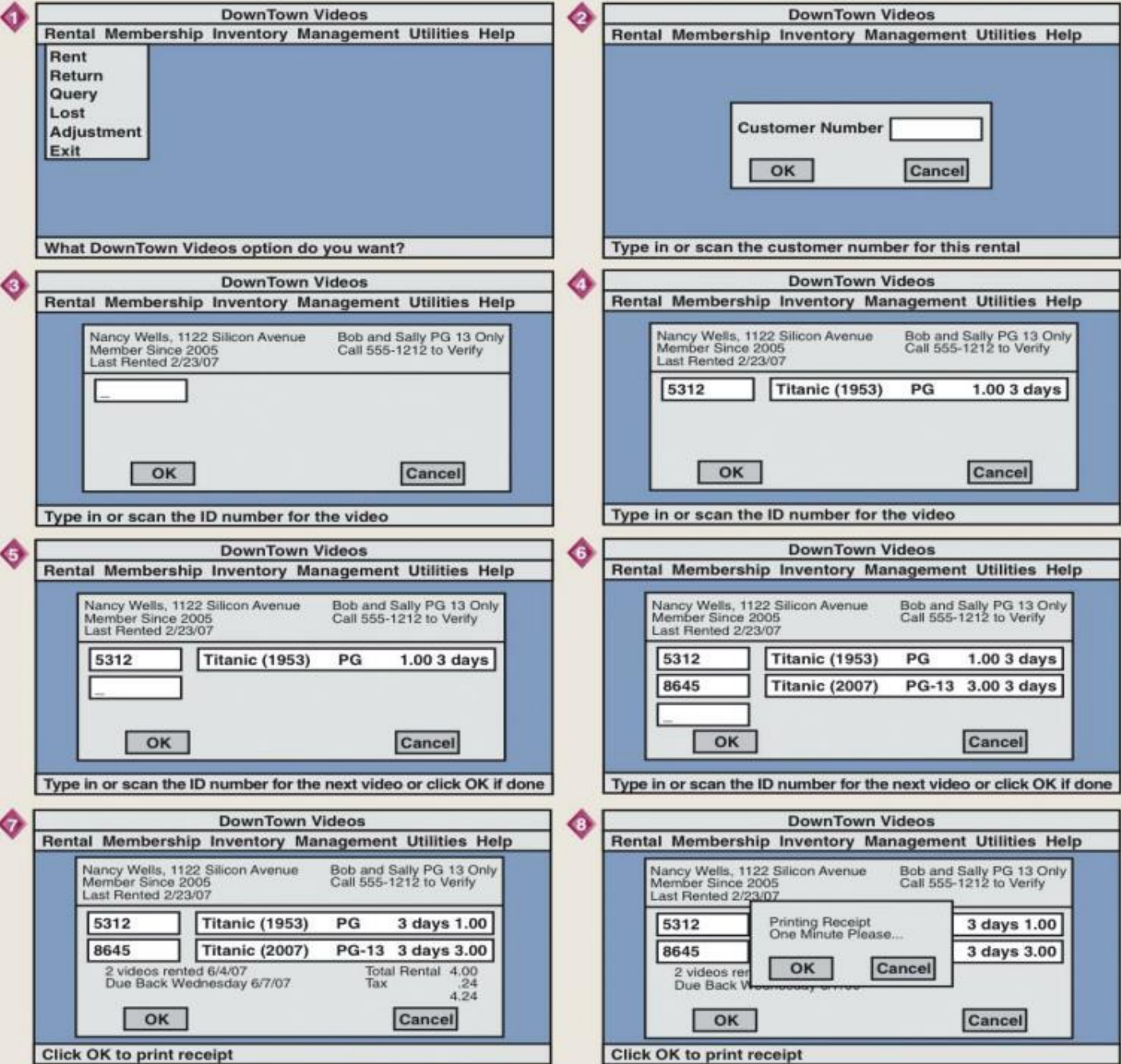
Storyboarding

Figure 7-2
Storyboarding
process

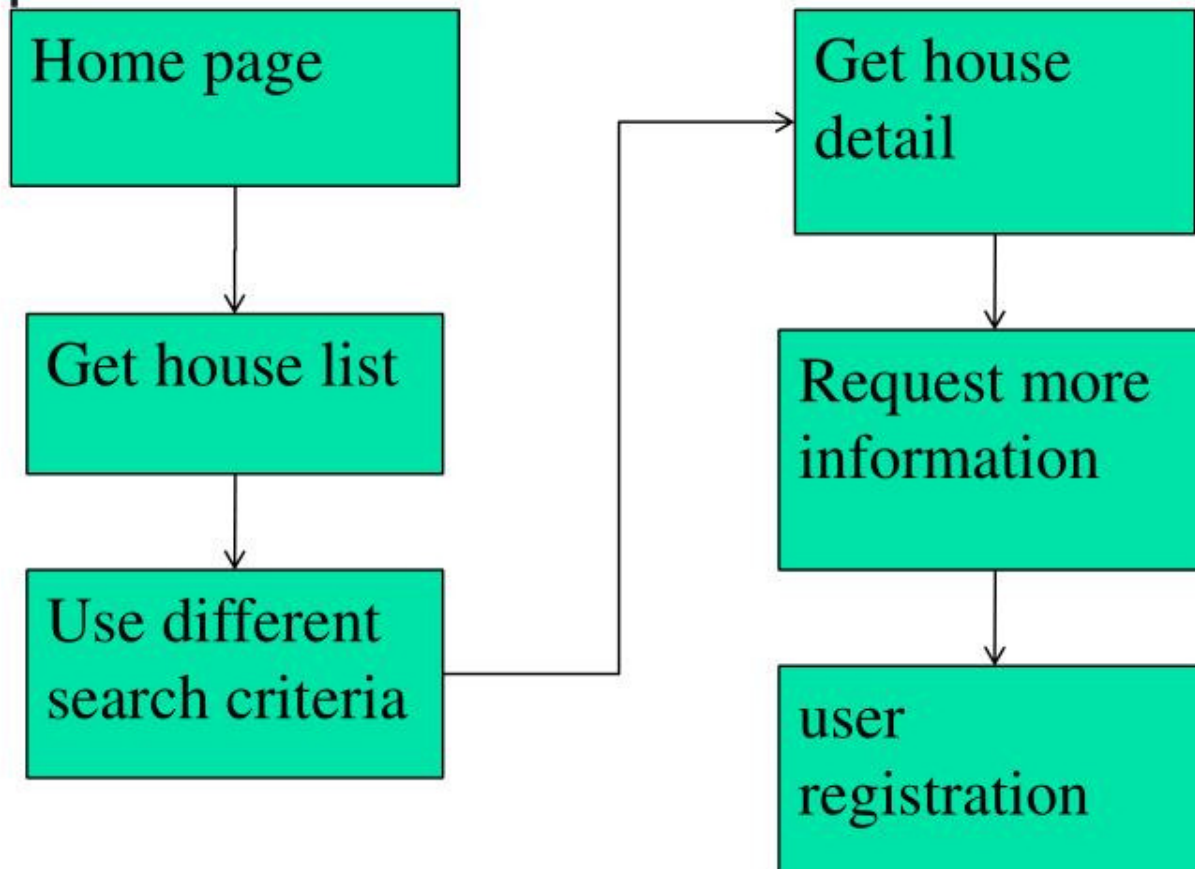




Storyboard
for *Rent
Videos*
Dialog



Story board for housing search





Our Mission

Westdale Housing has been serving the McMaster Student Housing community since 2000.

Our three areas of specialty are - Leasing, Daily operations, maintenance.

Investors trust our Property Management because we deliver results - Full tenancies, stable assets, appreciating investments.

Students - view off-campus housing listings. Click on any student house. See "Featured Properties" and "Properties for Lease" for available student houses.

We show student apartments, housing for international students and rooms for rent to students who are living the McMaster student life style.

4 and 5 BEDROOM STUDENT CONDO APARTMENT!!!

66-68 KING TO SEE PICTURES!!

WESTDALE HOUSING

Main Properties for Lease Lease Application Tenant Services Management Services About Us Contact Us

>> [Main](#)

! [Notify me when new properties are available!](#)

PROPERTIES FOR LEASE

Sort by: ☒ Price ☐ Sq. Ft. ☐ Bedrooms ▼ Descending



\$335 - Utilities Included Flexible
29 West 3rd St.
Hamilton, CA 90210
6 Bedrooms / 2 Baths / Large Sq. Ft.
[More Details...](#)



\$400 + Utilities 12 months
1164 King St. W.
Hamilton, CA 90210
7 Bedrooms / 2 Baths / Large Sq. Ft.
[More Details...](#)

WESTDALE HOUSING

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>> [Main](#) | [Homes for Lease](#)

[! Inquire about this property](#)

Residential Lease - Student Housing

**29 West 3rd St.
Hamilton, CA 90210**
[View map](#)

\$335 - Utilities Included
6 Bedrooms
2 Baths
Large square feet

Available: Now &/or April

Lease Term: Flexible

CLICK - "Additional Photos"

Beautiful Home close to campus. Perfect for a group of 6. Features include Hardwood floors, new paint, spacious bedrooms. \$335 per room - Utilities Included. Wireless Internet router included.

Pet Policy:

Please ask about our Pet Policy.



[Additional photos](#)



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>> [Main](#) | [Homes for Lease](#) | [Property Inquiry](#)

Please complete the following form to receive information regarding this property. We will contact you as soon as possible. Thank you.

Name:

Address:

City:

State:

Zip:

Daytime Telephone:

Evening Telephone:

Email Address:

Property Address: 29 West 3rd St. Hamilton, CA 90210

Off-Campus Resource Centre (OCRC)

Off-Campus Resource Centre Home



Off-Campus Resource Centre

+ LARGER FONT - SMALLER FONT PRINT

- Home
- Contact Us
- Students*
- Search for Rental Housing
- List/Manage an Ad
- Info for Students
- Landlords*
- Register for a New Account
- Log into your account
- List a New Ad(s)
- Manage your Ads
- View current Listings
- Info for Landlords
- Find a Realtor/Property Manager/Insurance Broker
- Resource Tools*
- Rental Accountability Program
- (Featured Listings)
- Info for Families/Faculty
- Info for Tenants & Landlords

Welcome, New to McMaster? Moving out of residence? Not sure how to find suitable housing? Look no further. The friendly staff in OCRC (Off-Campus Resource Centre) are here to help!

Search for available student housing in the McMaster Area. [\(click here\)](#). With our new listing service we have tonnes of great new features to help you find that perfect house. All maps and listings use



The OCRC serves primarily as a rental listing

Highlights

- **NOTICE TO ALL STUDENTS !**
 Education, self screening, early identification key to University's **H1N1** plan.
 - [Click here for further information.](#)
 - Click here the [Self-Screening Tool](#)
- **NOTICE to all LANDLORDS:**
 The action to mark your properties rented has been removed from the system for the time being. If your property has been rented, please



Special techniques in interface design

- Multimedia – voice, video, music, 3D image...
- Map and geographic information
- Virtual reality and animation
- Automatic phone answering system
- Mobile device user interface design

SitePal™

What does **your** site say?

Questions? [Contact Us](#)

| [MySitePal Login](#)

Demo

Customers

Features

Pricing

Developers



Enable your site with the power of voice

Sit back, relax and we will
show you how it works.



Watch Intro Movie

What is SitePal?

SitePal is an Internet service that allows you to create and add a fully customizable speaking avatar to your websites, Flash applications and emails. Programming skills not required.

Demo



Go Beyond Flat Text!

- Communicate with impact
- Enrich the user experience
- Engage visitors
- Stand out

Success stories

Sign up for a Free Trial!

Starting at
\$9.95
per month

Get started

FREE
15 DAY TRIAL

Case Study

Read how SitePal
virtual consultant
improved the
sales productivity
by 50%

Featured Customer



Photo-converted Avatars



Features

System components

MIDS process

Case studies

FAQ

Evaluate

Purchase

Support

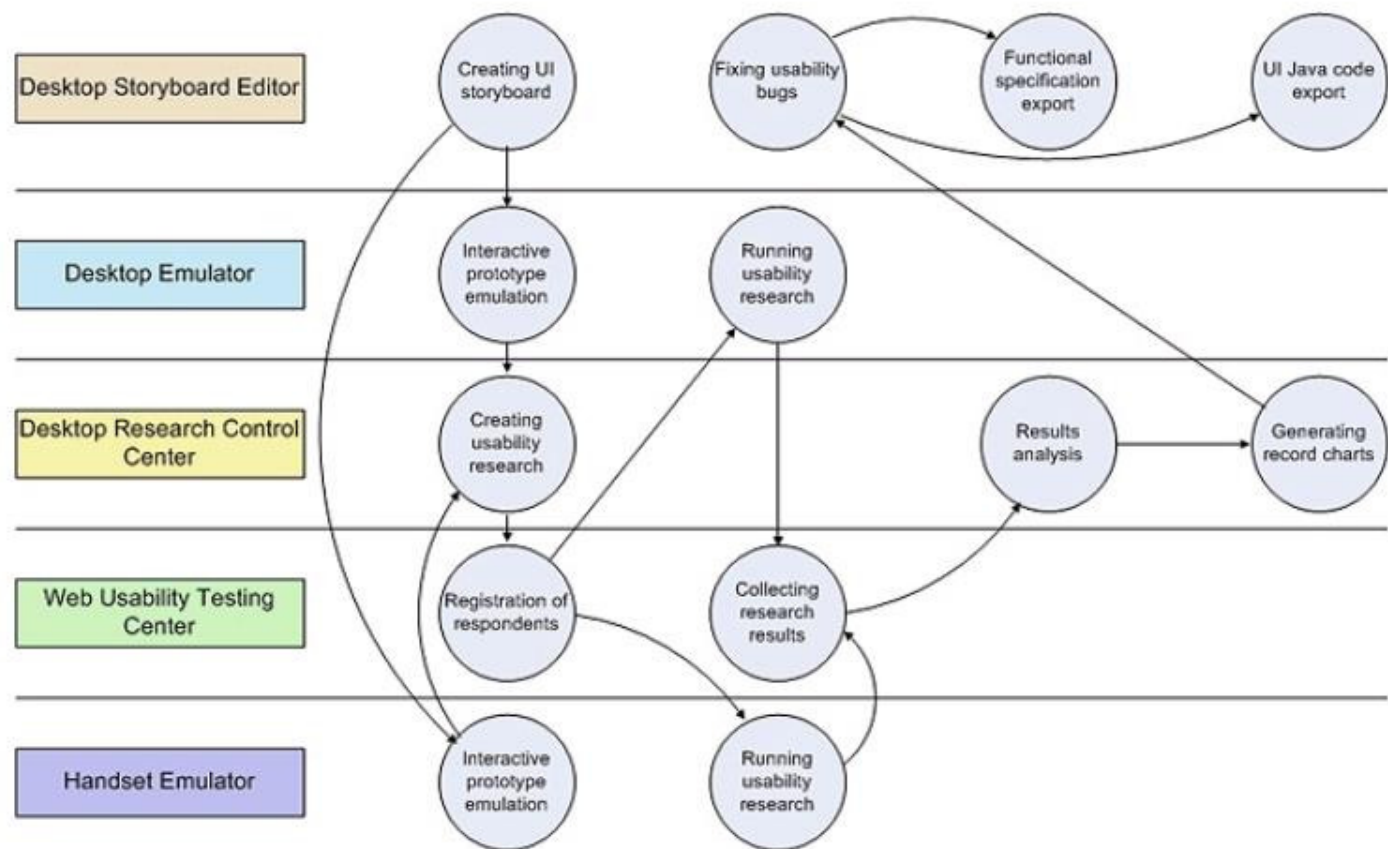
Feedback

News

Contacts

MIDS is designed to fully automate your UI prototyping and usability research process. MIDS workflow meets standard process requirements of cellular operators, handset manufacturers, content and service providers as well as design houses. You can pursue the following basic steps when employing MIDS for your process automation or we will customize the system to meet the specific needs of your company.

MIDS process



1. Designer creates a user interface storyboard employing rich library of controls and UI components of the MIDS Desktop Storyboard Editor.

2. Designer performs immediate testing of the prototype by pushing one button that converts the storyboard into



Summary

- User interface is everything user comes into contact with while using the system
 - Physically, perceptually, and conceptually
- To some users, user interface *is* the system
- User-centered design means
 - Focusing early on users and their work
 - Evaluating designs to ensure usability
 - Applying iterative development



Summary (continued)

- User interface is described with metaphors (desktop, document, dialog)
- Interface design guidelines and standards are available from many sources
- Dialog design starts with use cases and adds dialogs for integrity controls, user preferences, help
- OO approach provides UML models to document dialog designs, including sequence diagrams, activity diagrams, and class diagrams